



The combustion chemistry of ammonia and ammonia/hydrogen mixtures: A comprehensive chemical kinetic modeling study

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ARTICLE INFO

Keywords:

Ammonia

Hydrogen

Carbon-free combustion

Chemical kinetic model

Reaction mechanism

ABSTRACT

Ammonia (NH₃) is relatively less reactive compared to hydrocarbon fuels. Therefore, ammonia mixtures blended with hydrogen (H₂) have been shown to be a promising fuel for internal combustion engines. In this study, a detailed NH₃/H₂ chemical kinetic model is developed over a wide range of engine-relevant conditions and comprehensively validated to describe the combustion of NH₃/H₂ mixtures using available experimental literature data, including ignition delay times, laminar flame speeds and species concentration profiles. The new model captures very well the combustion properties of pure NH₃ and NH₃/H₂ mixtures at most conditions. By performing sensitivity and reaction path flux analyses the key reactions controlling fuel reactivity at high-temperature (≥ 1500 K) and low-to-intermediate temperature ($1000 \leq T \leq 1500$ K) regimes are identified. Moreover, the formation and consumption pathways of nitrogen oxides (NO_x) in NH₃/H₂ combustion at different conditions have also been investigated, which are found to be highly coupled to the underlying chemical reactions that dictate fuel reactivity. The kinetic data for the important reactions and species thermochemistry data used in our model are rigorously evaluated and are discussed in detail.

1. Introduction

Carbon dioxide (CO₂) emissions from fossil fuel combustion have increased almost exponentially since the 1960s and account for over 80% of total CO₂ emissions from different sources by the end of 2022 [1]. To mitigate CO₂ emissions caused by hydrocarbon fuel combustion, a transition from petroleum-based hydrocarbon fuels to carbon-free chemicals such as hydrogen or its carrier, ammonia (NH₃) is underway. Compared to liquefied hydrogen, liquefied ammonia has a higher volumetric hydrogen density and is easier to store and deposit. The pressure required to liquefy ammonia is 9.9 atm at 25.0 °C and 1.0 atm at −33.4 °C, similar to propane liquification [2]. In a recent report by the International Energy Agency (IEA), *The Future of Hydrogen* [3], ammonia is deemed an important sustainable energy carrier and has great potential to address environmental issues caused by transportation carbon emissions. Some recent reviews have demonstrated the feasibility and

prospect of ammonia as a fuel candidate for cleaner power [2,4–8]. However, ammonia is rather unreactive in its pure form, with relatively low laminar flame speeds of ~ 10 cm s^{−1}, an adiabatic flame temperature of ~ 1850 K and a narrow flammability limit of $\varphi = 0.63$ –1.40 at atmospheric pressure, which may lead to problems such as ignition failure. The co-firing strategy of blending ammonia with a more reactive combustion promoter can effectively facilitate the applications of ammonia in real engines [5]. Technical issues for ammonia/hydrogen binary fuel blends to be applied to gas turbines and internal combustion engines have been actively investigated and discussed [7–10]. On the other hand, excessive nitrogen oxide (NO_x) formation as a consequence of fuel-bound nitrogen is also a problem to be addressed [11]. To boost the efficient utilization of ammonia in practical engines and minimize the unwanted by-products of ammonia combustion, an in-depth understanding of the combustion chemistry of ammonia and ammonia/hydrogen mixtures is needed over a wide range of engine relevant

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<https://doi.org/10.1016/j.combustflame.2023.113239>

Received 24 August 2023; Received in revised form 28 November 2023; Accepted 1 December 2023

Available online 16 December 2023

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Table 1NH₃/H₂ IDT experimental data in the literature used to validate the combustion kinetic model developed in this work.

Fuel	T_5/T_C , K	p_5/p_C , atm	φ	Dilution	Ref.
Shock tube					
NH ₃	1560–2455	1.4, 11, 30	0.5, 1.0, 2.0	98% or 99% Ar	[12]
NH ₃	1100–1600	20, 40	0.5, 1.0, 2.0	In air	[29]
NH ₃ /H ₂ (0–70%)	1170–2000	1.2, 10	1.0	92% Ar	[13]
NH ₃ /H ₂ (50, 100%)	1438–1694	2.2	1.0	19% N ₂ + 78% Ar	[30]
RCM					
NH ₃ /H ₂ (0–20%)	960–1130	20, 40, 60	0.5, 1.0, 1.5, 2.0	70% Ar, 63% Ar + 7% N ₂	[31]
NH ₃ /H ₂ (0–25%)	1000–1100	43, 65	0.35	61% N ₂ + 11% Ar	[32]
NH ₃ /H ₂ (0–10%)	952–1210	20, 40, 60; 40–75	0.5, 1.0, 2.0, 3.0	70% Ar, 65% Ar + 10% N ₂ , 75% Ar	[33]
NH ₃ /H ₂ (0, 10, 25 %)	952–1408	30	0.5, 1.0, 1.5	85–93 % Ar	[34]
NH ₃ /N ₂ O (25%, 50%)					

Table 2NH₃/H₂ flame speed experimental data in the literature used to validate the combustion kinetic model developed in this work.

Mixture	T_i , K	p , atm	φ	Ref.
NH ₃ /air	298	1.05	0.9–1.2	[17]
NH ₃ /air	298–448	1	0.7–1.5	[18]
NH ₃ /O ₂ (21–45%)/N ₂	298	1, 2 & 5	0.6–1.5	[19]
NH ₃ /H ₂ (0–45%)/air	298	1	0.75–1.60	[20]
NH ₃ /H ₂ (0–60%)/air	298–473	1	0.8–1.4	[21]
NH ₃ /H ₂ (40, 60%)/air	298	1, 3, 5	0.6–1.6	[22]
NH ₃ (10–80% cracked)/air	298	1, 2, 5, 10	0.7–1.4	[23]
NH ₃ /H ₂ (0–30%)/air	298–473	1, 3; 5–10	0.8–1.4	[24]
NH ₃ /O ₂ (21–30%)/N ₂ /He				
NH ₃ /NO (50%)/N ₂ (50%)	298	1	1.1–1.9	[25]
NH ₃ /O ₂ (30%)/N ₂ /He	298	1–7	1.0, 1.2, 1.4	[26]
NH ₃ /H ₂ (40%)/air	298	1, 5	0.8–1.8	[27]
NH ₃ /H ₂ (0–80%)/air	298	1	0.6–1.4	[28]

thermodynamic conditions. Fundamental combustion experiments of ammonia and ammonia/hydrogen mixtures provide valuable information of how global reactivity of the fuel mixtures and the generation and destruction of chemical species of interest depend on temperature (T), pressure (p), equivalence ratio (φ), blending ratio, etc. The experimental data also serves as validation targets that are necessary to develop an accurate predictive combustion kinetic model. A survey of fundamental combustion experiments of ammonia and ammonia/hydrogen mixtures in the literature is performed in this work, including measurements of ignition delay times (IDTs), laminar flame speeds and species profiles.

Mathieu and Petersen [12] used a shock tube to measure IDTs of NH₃/O₂/Ar mixtures at high temperatures in the range 1560–2455 K, at pressures of approximately 1.4, 11 and 30 atm, and at equivalence ratios

of 0.5, 1.0 and 2.0. Chen et al. [13] investigated the effect of H₂ addition on the autoignition characteristics of NH₃ in a shock tube for stoichiometric NH₃/H₂/O₂/Ar mixtures in the temperature range 1020–1945 K, at pressures of 1.2 and 10 atm, for H₂ mole fractions of 0%, 5%, 30% and 70% in the NH₃/H₂ mixtures. Davidson et al. [14] measured the time histories of $\dot{N}H$ and $\dot{N}H_2$ radicals formed during the pyrolysis of NH₃ diluted by Ar at ~ 1 atm and at temperatures above 2300 K. Alturaifi et al. studied the time evolutions of NH₃ [15] and N₂O [16] concentrations in a shock tube at near-atmospheric pressure during the pyrolysis of NH₃/H₂ mixtures in Ar in the temperature range 2096–3007 K, and they also studied the oxidation of NH₃/O₂/Ar mixtures in the temperature range 1829–2198 K, respectively. There have been a number of experimental laminar flame speed (S_L) studies of NH₃/H₂/air mixtures by different research groups [17–28], at equivalence ratios mostly in the range 0.8–1.4, at unburnt gas temperatures of 298–473 K, and at pressures of 1 atm to 10 atm. The H₂ ratio in the fuel varied from 0 to 60 mol % with multiple O₂ fractions in the oxidizer ranging from 21 to 45 mol %. The high-temperature experimental data of NH₃/H₂ and their corresponding experimental conditions are summarized in Tables 1 and 2.

In the low-to-intermediate temperature regimes, Shu et al. [29] measured IDTs in a shock tube for NH₃/air mixtures in the temperature range 1180–1580 K, at $p = 20$ and 40 bar, and at equivalence ratios of 0.5, 1.0 and 2.0. He et al. [31] and Dai et al. [33] both used rapid compression machines (RCMs) to study the autoignition of NH₃/H₂/N₂/Ar mixtures at similar conditions, and reported IDTs in the temperature range 1020–1220 K, at pressures of 40 and 60 bar, and at equivalence ratios of 0.5–3.0 for pure NH₃. The conditions varied to the temperature range 950–1180 K, at pressures in the range 20–60 bar, and at equivalence ratios of 0.5–2.0 for NH₃/H₂ mixtures with 1–20 mol % H₂ in the fuel. Pochet et al. [32] obtained IDTs at fuel-lean conditions (φ

Table 3NH₃/H₂ speciation experimental data in the literature used to validate the combustion kinetic model developed in this work.

Fuel	T/T_i , K	p , atm	φ	Dilution	τ , s	Ref.
JSR						
NH ₃ /NO (0–50%)	1100–1460	1.00	0.1–2.0	99% N ₂	0.1, 0.2	[35]
NH ₃	500–1200	1.05	0.010, 0.019	97% He	1.5	[36]
NH ₃ /H ₂ (0–70%)	800–1280	1.00	0.25, 1.00	99% N ₂	1.0	[37]
NH ₃	700–1200	1.00	0.1, 0.5, 1.0	83–97% Ar	3.0	[38]
NH ₃ /H ₂ (0, 30%)	800–1300	1.00	0.6, 1.0, 1.5	99% Ar	1.0	[39]
Shock tube						
NH ₃	2000–3200	0.80–1.10	∞	99% Ar	/	[14]
NH ₃ /H ₂ (0, 83%)	2100–3000	~ 1.00	∞	99% Ar	/	[15]
NH ₃	1829–2198	~ 1.20	0.5, 1.0, 1.8	97% Ar	/	[16]
Flow reactor						
NH ₃	700–1000	30, 100	0.200	96% N ₂	T -dependent	[40]
NH ₃	1300–2000	1.25	0.375	99% He	~ 0.05	[36]
NH ₃	875–1450	1.00	0.046– ∞	99% N ₂	T -dependent	[41]
NH ₃ /H ₂ (0–100%)	875–1800	1.25	0.60–0.78	99.7% He	~ 0.05	[42]
Flame structures						
NH ₃ /H ₂ (0, 50%)	368	1.00	0.8, 1.0, 1.2	$\sim 70\%$ Ar	/	[43]
NH ₃ /H ₂ (50%)	368	4.00, 6.00	0.8, 1.0, 1.2	$\sim 70\%$ Ar	/	[44]

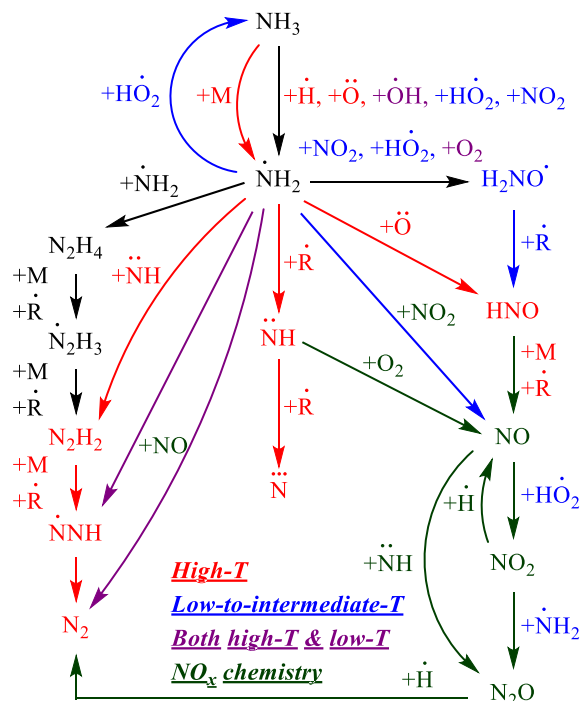


Fig. 1. Generalized reaction scheme for ammonia oxidation.

= 0.2–0.5) and temperatures and pressures of 1000–1090 K and 37–75 bar, respectively, for $\text{NH}_3/\text{O}_2/\text{N}_2/\text{Ar}$ mixtures in an RCM. The low-to-intermediate-temperature NH_3/H_2 IDT experimental data are summarized in Table 1. Speciation of NH_3 oxidation were probed by Stagni et al. [36] in a jet-stirred reactor (JSR) at super-lean conditions of $\varphi = 0.01$ –0.02 and in a plug flow reactor (PFR) at $\varphi = 0.35$. More recently, Zhang et al. [37] reported speciation data for NH_3/H_2 mixtures at atmospheric pressure and $\varphi = 0.25$ and 1.00 with H_2 content varying from 0 to 70 mol % in a JSR, and reported concentration profiles for NH_3 , H_2O , NO and N_2O . Tang et al. [38] measured the concentration profiles for similar species with N_2 , H_2 and NO_2 additionally for the oxidation of pure NH_3 in a JSR at $\varphi = 0.1$, 0.5 and 1, and at $p = 1$ atm. The available NH_3/H_2 speciation experimental data in the literature with their experimental conditions are summarized in Table 3.

To fully elicit the underlying chemistry of ammonia and ammonia/hydrogen combustion, over the last two decades continuous efforts have been devoted to the development of detailed combustion kinetic models. To date, a number of ammonia combustion kinetic models have been published, among which the recent models developed by Glarborg et al. [45], Otomo et al. [46], Stagni et al. [36] and Zhang et al. [37] are generally validated against more experimental data and show a better performance compared to the others [12,19,47,48]. However, limitations in the predictive capability of these models for the combustion properties of ammonia and ammonia/hydrogen mixtures are still evident, particularly over a wide range of engine-relevant conditions. For example, Shrestha et al. [24] tested several NH_3 kinetic models against their experimentally measured NH_3/H_2 laminar flame speeds at $p = 1$ bar and $T_i = 473$ K, and showed that the Glarborg et al. model significantly over-estimated and the Otomo et al. model under-estimated their S_L for mixtures with > 5 mol % H_2 . The Stagni et al. and Zhang et al. models generally predict the S_L of NH_3/H_2 mixtures well, but they both tend to over-estimate the high-temperature IDTs of pure ammonia reported by Mathieu and Petersen [12] and Chen et al. [13]. In addition, the Stagni model significantly over-estimates the yields of N_2O in pure NH_3 and NH_3/H_2 oxidation [16,37], and none of these models can satisfactorily capture the low- T IDTs measured in RCMs for both NH_3 and NH_3/H_2 mixtures. In this context, the aim of this study is to thoroughly understand the detailed chemistry that governs the reactivity of

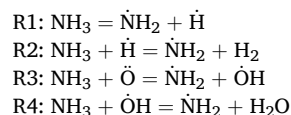
pure NH_3 and NH_3/H_2 mixtures, and the associated key reaction steps controlling the formation of NO_x in this system. In this work, a sophisticated NH_3 combustion kinetic model has been developed and comprehensively validated against the available data from fundamental combustion experiments of NH_3 and NH_3/H_2 mixtures in the literature. The combustion chemistry of NH_3 at both high- and low-to-intermediate temperature regimes, the important reaction pathways for NO_x formation and consumption, and the effect of H_2 blending on the chemistry will be discussed in detail.

2. Chemical kinetic mechanism development

A detailed chemical kinetic model for NH_3/H_2 combustion has been developed here. The model is constructed based on an well-validated in-house H_2/O_2 sub-mechanism from Galway updated from NUIGMech1.3 [49]. Notably, the chain branching product channel of $\text{HO}_2 + \text{HO}_2 = \text{OH} + \text{OH} + \text{O}_2$ reported recently by Klippenstein et al. [50] is included in the sub-mechanism, and the rate constants for $\text{H}_2 + \text{O} = \text{H} + \text{OH}$ and $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$ are also updated as illustrated in Figs. S1–S3 in the Supplementary Material (SM). The reaction subset of excited hydroxyl radical (OH^*) proposed by Kathrotia et al. [51] is adopted in the model to characterize OH^* formation and consumption. The high-temperature and low-to-intermediate-temperature reactions of NH_3 oxidation and NO_x formation and consumption reactions coupled with NH_3 oxidation are added to the model with their generalized scheme illustrated in Fig. 1. Since pure NH_3 is rather un-reactive as a fuel and its oxidation onset temperature is approximately 1200 K at atmospheric pressure [35–37,39], we define the low-to-intermediate-temperature range for NH_3 combustion to be 1000–1500 K and the high-temperature combustion regime to be at temperatures above 1500 K. The important reactions in these regimes and their rate constants are discussed here. The calculated thermodynamic data by Bugler et al. [52] are adopted for most nitrogen species in the model, while for NH_3 and NNH the data from Klippenstein et al. [47] is used. The standard enthalpies of formation ($\Delta_f H_{298}^\circ$) of NH_3 and NNH from Klippenstein et al. (with the enthalpy of the latter decreased by 0.5 kcal mol $^{-1}$ to ensure improved predictions of NO formation in H_2 flames) (–11.0 and 59.2 kcal mol $^{-1}$, respectively) [47] are closer to the benchmark values in the ATcT database (–10.9 $\pm$ 0.007 and 59.6 $\pm$ 0.1 kcal mol $^{-1}$, respectively) [53] compared to those calculated by Bugler et al. (–10.4 and 60.3 kcal mol $^{-1}$, respectively) [52]. The transport data are taken from NUIGMech1.3 for all species in the model in order to simulate one-dimensional flames where temperature and concentration gradients need to be considered. The available experimental data obtained from fundamental combustion experiments of NH_3/H_2 in the literature, listed in Tables 1–3, are used to validate the detailed ammonia combustion kinetic model developed in this study. The model validation results for all these targets are provided in the text and as SM.

2.1. High-temperature chemistry of NH_3 oxidation

At temperatures above 1500 K fuel oxidation can be initiated by unimolecular decomposition and H-atom abstraction reactions, R1–R4, as illustrated in Fig. 1.



Low-pressure limit (LPL) rate constants for NH_3 unimolecular decomposition (R1) were reported by Davidson et al. [14] through the analysis of literature measurements, and determined by Altinay and Macdonald [54] through their measurements for its reverse process, respectively. More recently, Stagni et al. [36] employed an ab initio

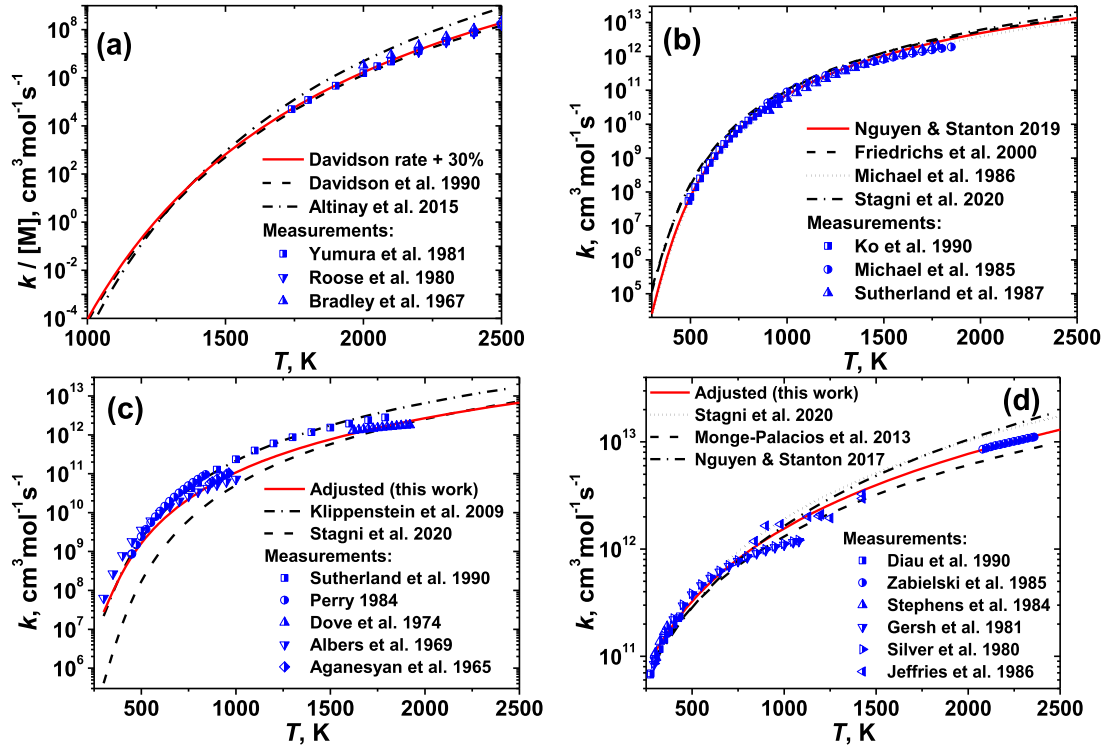


Fig. 2. Rate constants for (a) $\text{NH}_3 + \dot{\text{H}} = \dot{\text{N}}\text{H}_2 + \text{H}_2$ [14,36,54] at low pressure limit (LPL) with experimental measurements from [69–71], (b) $\text{NH}_3 + \dot{\text{H}} = \dot{\text{N}}\text{H}_2 + \text{H}_2$ [36,55,72,73] with experimental measurements from [56,74,75], (c) $\text{NH}_3 + \dot{\text{O}} = \dot{\text{N}}\text{H}_2 + \dot{\text{O}}\text{H}$ [36,57] with experimental measurements from [58–60,76,77], and (d) $\text{NH}_3 + \dot{\text{O}}\text{H} = \dot{\text{N}}\text{H}_2 + \text{H}_2\text{O}$ [36,61,62] with experimental measurements from [63–68].

transition state theory (TST) based master equation approach (AITSTME), including variational transition state theory (VTST) treatments, to calculate the pressure-dependent rate constants for R1 and the rate constants for H-atom abstraction reactions from NH_3 by $\dot{\text{H}}$, $\dot{\text{O}}$, $\dot{\text{O}}\text{H}$, $\text{H}\dot{\text{O}}_2$ and O_2 . As shown in Fig. 2(a), the LPL rate constants recommended by Davidson et al. match the measured data points quite well, while those from Altinay et al. [54] are somewhat faster and are 2.7 times higher than that reported by Davidson at 2000 K. A comparison of the rate constants for R1 at 1 atm is shown in Fig. S4, where the Altinay rate constant is 2.2 times higher than that reported by Davidson, while that from Stagni is 1.7 times faster than that of Davidson et al. at 2000 K. In this work, we employ the rate constants from Davidson et al. [14] but increased the frequency factor by 30% to obtain better agreement between model prediction and experimentally measured fuel concentration profiles for NH_3/H_2 pyrolysis measured by Alturaifi et al. [15] at approximately 1 atm in a shock tube.

After the radical pool is formed, NH_3 is mainly consumed via H-atom abstraction reactions by $\dot{\text{H}}$, $\dot{\text{O}}$ and $\dot{\text{O}}\text{H}$ radicals at high temperatures as shown in Fig. 1. For H-atom abstraction reaction from NH_3 by $\dot{\text{H}}$ (R2), Nguyen and Stanton [55] performed ab initio calculations where the vibrational ('umbrella') mode of NH_3 was treated as a one-dimensional hindered internal inversion. The Nguyen and Stanton rates are adopted in our model for R2, which are in good agreement with experimental measurements at 500–1800 K and lie in between the rate constants reported from Stagni et al. [36] and Michael et al. [56], Fig. 2(b). For $\text{NH}_3 + \dot{\text{O}} = \dot{\text{N}}\text{H}_2 + \dot{\text{O}}\text{H}$ (R3), the rate constants calculated by Stagni et al. [36] are a slightly slower than those calculated by Klippenstein et al. [57], with a deviation of around a factor of three between them at temperatures above 1000 K. In this work, we adjust the Klippenstein rates for R3 to match the experimentally measured data by Dove et al. [58] at high temperatures while maintaining the good agreement with the rate constants measured by Albers et al. [59] and Aganesyan et al. [60] at low temperatures, Fig. 2(c). For $\text{NH}_3 + \dot{\text{O}}\text{H} = \dot{\text{N}}\text{H}_2 + \text{H}_2\text{O}$ (R4), a deviation of about 80% is observed between the calculated rate constants

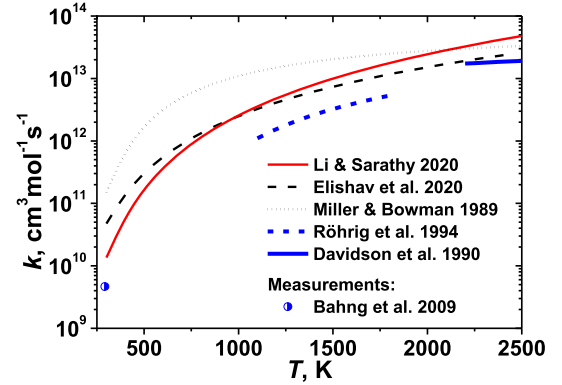
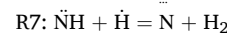
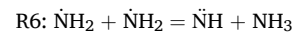
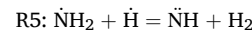


Fig. 3. Rate constants for $\dot{\text{N}}\text{H}_2 + \dot{\text{H}} = \dot{\text{N}}\text{H} + \text{H}_2$ [78–81] with experimental measurements from [14,82].

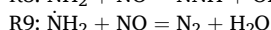
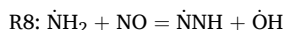
from Monge-Palacios et al. [61] and those calculated by Stagni et al. [36] and Nguyen and Stanton [62] at 2000 K, Fig. 2(d). We have adjusted the Stagni rates to match the experimental measured rate constants from Zabelski and Seery [63] at high temperatures and those from Jeffries and Smith [64] at intermediate temperatures, while maintaining good agreement with the experimental measurements at low temperatures [65–68].

Amino ($\dot{\text{N}}\text{H}_2$) radicals can undergo H-atom abstraction reactions primarily with $\dot{\text{H}}$ and itself to produce imidogen ($\dot{\text{N}}\text{H}$) radicals, and the H atom in the $\dot{\text{N}}\text{H}$ radical will mainly be abstracted by $\dot{\text{H}}$ atoms in the radical pool, as follows:



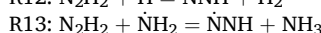
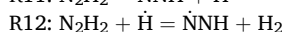
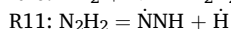
The H-atom abstraction reactions of NH_3 , $\dot{\text{N}}\text{H}_2$ and $\dot{\text{N}}\text{H}$ with $\dot{\text{H}}$ atoms (R2, R5 and R7) are important in inhibiting fuel reactivity at high-temperatures since they consume $\dot{\text{H}}$ atoms and compete with the important $\dot{\text{H}} + \text{O}_2 = \ddot{\text{O}} + \dot{\text{O}}\text{H}$ chain branching reaction. Li and Sarathy [79] recently performed TST calculation based on CCSD(T)/cc-pVXZ ($X = \text{Q}, \text{T}$)/M06-2X/6-311++G(d,p) energy barrier and rovibrational properties to obtained the rate constant for H-atom abstraction from $\dot{\text{N}}\text{H}_2$ by $\dot{\text{H}}$ atoms (R5). Their calculated rate constant is closer to the measured data at 293 K by Bahng and Macdonald [82] but it is about a factor of two faster than the refitted rate by Elishav et al. [78] based on literature review at temperature around 2000 K, Fig. 3. Considering the high level calculation method employed and a better model performance in predicting NH_3/H_2 flame speeds resulted from this faster rate constant in the high-temperature regime, the Li and Sarathy [79] rate constant for R5 is adopted. For the reaction system of $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H}_2$, Klippenstein et al. [57] performed ab initio calculations to obtain individual rate constants for each reaction pathway on its detailed potential energy surfaces, and their calculated rate constants are adopted in the model for R6 and other product channels of $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H}_2$. For $\dot{\text{N}}\text{H} + \dot{\text{H}} = \text{N} + \text{H}_2$ (R7), the calculated rate constants from Yao et al. [83] were used in the model which are within 20% with the only measured data of $\sim 3.0 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ reported by Morley [84] at 1790 K.

Nitric oxide (NO) addition reactions to $\dot{\text{N}}\text{H}_2$ radical are important in governing NH_3/H_2 fuel reactivity under both high-temperature and low-to-intermediate-temperature conditions. The $\dot{\text{N}}\text{H}_2 + \text{NO}$ reaction system has two product channels, namely,



Where R8 produces radical species and thus this reaction enhances fuel reactivity, whereas R9 produces two stable molecules and thus decreases reactivity. Therefore, the branching ratio of these two reaction pathways needs to be accurately described in the model. Glarborg et al. [45] extensively reviewed the experimental and theoretical investigations of $\dot{\text{N}}\text{H}_2 + \text{NO}$. They evaluated rate constants from the calculations of Klippenstein et al. [47,85] and the shock tube experimental-based work of Song et al. [86] by comparing them with available literature experimental data. Fig. S5 of the SM shows that the rate constants from Song provide the best fits to the experimental measurements for both the total rate constant of the $\dot{\text{N}}\text{H}_2 + \text{NO}$ [87–91] channel and the branching ratio of $\dot{\text{N}}\text{H}_2 + \text{NO} = \dot{\text{N}}\text{NH} + \dot{\text{O}}\text{H}$ [86,91–94] in the temperature range 300–2000 K. Therefore, we selected the rate constants from Song et al. [86] for the $\dot{\text{N}}\text{H}_2 + \text{NO}$ pathway in our model.

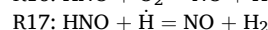
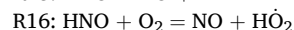
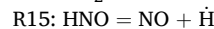
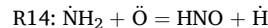
Diazine (N_2H_2) is an important intermediate formed at high-temperature conditions. Fig. 1 shows that it can be produced from the self-recombination of $\dot{\text{N}}\text{H}_2$ and subsequent dehydrogenation of the adduct N_2H_4 molecules through unimolecular decomposition and H-atom abstraction reactions. Alternatively, it can be produced by the reaction of $\dot{\text{N}}\text{H}_2$ with $\dot{\text{N}}\text{H}$. The N_2H_2 molecules will then undergo unimolecular decomposition and H-atom abstraction reactions mainly with $\dot{\text{H}}$ and $\dot{\text{N}}\text{H}_2$ in the radical pool that compete with the decomposition, as follows:



Elishav et al. [78] reviewed the available literature rate constants for $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H} = \text{N}_2\text{H}_2 + \dot{\text{H}}$ (R10), where experimental measurements were limited to approximately room temperature for the total rate constants of $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H}$. The rate constants for R10 calculated by Klippenstein et al. [57] and those from the earlier fitting of that reported by Davidson et al.

[14] are within 75% in the temperature range 300–2500 K, with the Klippenstein rate constant being slightly faster at high temperatures, as illustrated in Fig. 4. In this work, we adopt the Davidson et al. rate constant [14] for R10. For the N_2H_2 unimolecular decomposition reaction, R11, the only available rate constants in the literature are those calculated by Dean and Bozzelli [95] using QRRK estimates for pressures in the range of 0.1–10 atm. In order to extrapolate this rate constant to pressures higher than 10 atm, we obtained a LPL rate constants expression for R11 as $k/[M] = 1.41 \times 10^{43} T^{-7.60} \exp(-74.77 \text{ kcal}/RT) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ by fitting to the Dean and Bozzelli rate constants at 0.1–10 atm. The fitted rate constants are within 20% of the Dean and Bozzelli values at temperatures above 1200 K and at $p = 1$ and 10 atm, as shown in Fig. S6(a). Figure S6(b) shows that the calculated rate constants for $\text{N}_2\text{H}_2 + \dot{\text{H}} = \dot{\text{N}}\text{NH} + \text{H}_2$ (R12) by Li and Sarathy [79], Zheng et al. [96] and Linder et al. [97] vary considerably (by up to an order of magnitude), with the Li and Sarathy rate constant being the fastest. This difference may be largely due to the different calculated energy barriers for R12, where an energy barrier of $2.95 \text{ kcal mol}^{-1}$ was reported by Zheng et al. using the M08-SO/ma-TZVP density functional method while a lower energy barrier of $1.95 \text{ kcal mol}^{-1}$ was obtained by Li and Sarathy [79] based on a coupled cluster CCSD(T)/cc-pVXZ ($X = \text{Q}, \text{T}$) calculation. An adjusted rate constant falling between these data is used in our model to better capture the high- T combustion properties of pure NH_3 . The rate constants for $\text{N}_2\text{H}_2 + \dot{\text{N}}\text{H}_2 = \dot{\text{N}}\text{NH} + \text{NH}_3$ (R13) are shown in Fig. S6(c), with the more recently calculated rate constant by Diévar et al. [98] employing a higher level of theory used in the model and favored over the rate constant calculated earlier by Linder et al. [97].

For NH_3/H_2 mixtures in the high-temperature regime, particularly in flames where $\ddot{\text{O}}$ atoms are abundant, HNO molecules can be produced through $\dot{\text{N}}\text{H}_2$ addition to $\ddot{\text{O}}$, with the HNO product consumed by unimolecular decomposition and H-atom abstraction reactions primarily with O_2 , $\dot{\text{H}}$ and $\dot{\text{N}}\text{H}_2$ as follows:



Klippenstein et al. [101] recently investigated the potential energy surfaces (PEs) of the $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}}$ reaction system, using the state-of-the-art ANL1//CCSD(T)-F12/cc-pVQ,T-F12 level of theory, and obtained rate constants from a combined variable reaction coordinate transition state theory (VRC-TST) and master equation calculation. The formations of $\text{HNO} + \dot{\text{H}}$ (R14) and $\dot{\text{N}}\text{H} + \dot{\text{O}}\text{H}$ were found to be the major reaction channels of $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}}$ in the high- T regime, while a third reaction pathway, $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}} = \text{NO} + \text{H}_2$, accounts for $\sim 20\%$ of the total rate

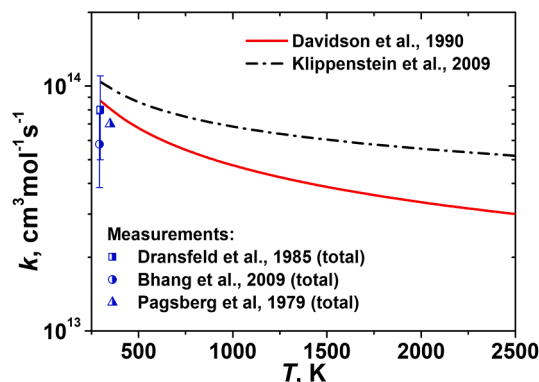


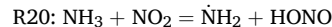
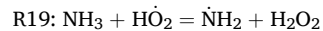
Fig. 4. Rate constants for $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H} = \text{N}_2\text{H}_2 + \dot{\text{H}}$ [14,57] with experimental measurements for $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H} = \text{products}$ from [82,99,100].

constant. Their calculated rate constants are adopted in our model. For $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}} = \text{HNO} + \dot{\text{H}}$ (R14) and $\text{NH}_2 + \ddot{\text{O}} = \dot{\text{N}}\text{H} + \dot{\text{O}}\text{H}$, the Klippenstein rate constants are plotted together with available rate expressions and experimental data in Fig. 5. The Klippenstein rate constant for R14 is about a factor of two slower than that calculated by Bozzelli and Dean [102] and Sumathi et al. [103] and the limited experimental data at temperatures below 500 K [99,104], Fig. 5(a). At $T > 1200$ K where $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}}$ reactions become important, the slower Klippenstein rate constant for R14 results in good model predictions of both fuel reactivity and NO concentrations for NH_3/H_2 mixture combustion. For $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}} = \dot{\text{N}}\text{H} + \dot{\text{O}}\text{H}$, which proceeds through both direct a H-atom abstraction and a chemically activated $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}} \rightarrow [\text{NH}_2\text{O}]^* \rightarrow \dot{\text{N}}\text{H} + \dot{\text{O}}\text{H}$ pathway, the Klippenstein rate constant and earlier ones from Dean and Bozzelli [95] and Duan and Page [105] are within 50% of one another at $T > 1500$ K, Fig. 5(b). Stagni et al. [36] calculated the pressure-dependent rate constants for $\text{HNO} = \text{NO} + \dot{\text{H}}$ (R15) and demonstrated good agreement between their calculated LPL rate constants for the reverse process of R15 with experimental measurements in the literature. Hence, we have adopted the Stagni calculation for R15 in our model. For H-atom abstraction reaction of HNO by O_2 (R16) a rate constants expression of $2.0 \times 10^{13} \exp(-14.00 \text{ kcal}/RT) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was adopted in the model, which is from Dean and Bozzelli [95] estimations by using the “direct hydrogen transfer” (DHT) approach with E_a decreased by $1.9 \text{ kcal mol}^{-1}$ to match the experimental measurements of Bryukov et al. [106] for $\text{HNO} + \text{O}_2 = \text{products}$ at 296–421 K. The reaction $\text{HNO} + \dot{\text{H}} = \text{NO} + \text{H}_2$ (R17) proceeds through both direct H-atom abstraction and chemically activated reaction pathways. Klippenstein et al. reported the calculated rate constant for R17 in their recent study [101], which is also on the $[\text{NH}_2\text{O}]$ PESs, with the direct H-atom abstraction transition state treated based on microcanonical VTST. As shown in Fig. S7(a), the Klippenstein rate constant is about a factor of two to three faster than the ones calculated by Nguyen et al. [107] and Soto et al. [108], while that from Kovács [109], determined by fitting a mechanism, is an order of magnitude slower than the calculated ones at high temperatures. We adopted the Klippenstein rate constant [101] for R17 in our model. For H-atom abstraction from HNO by $\dot{\text{N}}\text{H}_2$ (R18), Xu and Lin [110] and Mebel et al. [111] both calculated the rate constants for its reverse process ($\text{NH}_3 + \text{NO} = \dot{\text{N}}\text{H}_2 + \text{HNO}$), at the CCSD(T)/6–311+G(3df, 2p), G1 and G2 levels of theory. The Xu and Lin rates are approximately an order of magnitude faster than the Mebel rates, Fig. S7(b), and the former were used in the model for R18 since Xu and Lin employed a higher level of theory in their calculations.

2.2. Low-to-intermediate-temperature chemistry of NH_3 oxidation

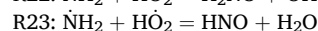
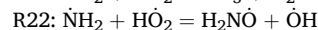
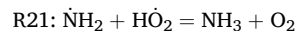
In the combustion of NH_3/H_2 mixtures at low-to-intermediate-temperatures (1100–1500 K) and at pressures above 30 atm, the

formation of hydroperoxyl (HO_2) radicals and nitrogen oxide (NO_2) molecules are favored through $\dot{\text{H}} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$ and $\text{NO} + \text{HO}_2 = \text{NO}_2 + \dot{\text{O}}\text{H}$, respectively. The fuel molecules can react with HO_2 and NO_2 through the following H-atom abstraction reactions:



Recently, Stagni et al. [36] calculated the rate constant for $\text{NH}_3 + \text{HO}_2 = \dot{\text{N}}\text{H}_2 + \text{H}_2\text{O}_2$ (R19) and we use this value in our mechanism. For the reaction $\text{NH}_3 + \text{NO}_2$, the product channel leading to the formation of $\dot{\text{N}}\text{H}_2 + \text{HONO}$ (R20) predominates over the channel forming $\dot{\text{N}}\text{H}_2 + \text{HNO}_2$. Xu and Lin [113] performed ab initio calculations for R20 and showed good agreement with their calculated total rate constants for $\text{NH}_3 + \text{NO}_2$ and available experimental literature measurements. Hence, we have adopted the Xu and Lin rate constant for R20.

At low temperatures, $\dot{\text{N}}\text{H}_2$ can undergo barrierless bi-radical addition reactions with HO_2 and enter into a multi-well, multi-channel reaction system or can directly abstract the H atom from HO_2 . The product channels for the reactions of $\dot{\text{N}}\text{H}_2$ with HO_2 can be described as follows:



This reaction system is important in controlling fuel reactivity at low temperatures (≤ 1200 K) because the chain terminating product channel R21 competes with R22, which generates highly reactive $\dot{\text{O}}\text{H}$ and nitroxide ($\text{H}_2\text{N}\dot{\text{O}}$) radicals. Klippenstein and Glarborg [114] recently carried out high-level ab initio calculations on the $\dot{\text{N}}\text{H}_2 + \text{HO}_2$ reaction system. They explored the PESs using the ANLO [121] method based on the molecular geometries obtained at the CCSD(T)-F12/cc-pVTZ-F12 level of theory and employed VRC-TST to calculate the bottle-neck reactive fluxes for the barrierless entrance reaction channels. Their calculations indicated very little pressure dependence of the $\dot{\text{N}}\text{H}_2 + \text{HO}_2$ reactions, and they concluded that at temperatures in the range 300–2000 K, the formation of $\text{NH}_3 + \text{O}_2$ (R21) is the dominant product channel ($\sim 80\%$), while the formation of $\text{H}_2\text{N}\dot{\text{O}} + \dot{\text{O}}\text{H}$ (R22) accounts for $\sim 20\%$ of the flux with the formation of $\text{HNO} + \text{H}_2\text{O}$ (R23) being marginal. Rate constants for R21 were also reported in the calculations by Chavarrio Cañas et al. [115], and the reverse rate constants for R21 were calculated by Stagni et al. [36]. As shown in Fig. 6(a), the rate constants from Klippenstein and Glarborg [114] for R21 and that from Stagni (calculated from reverse rate constants through microscopic reversibility) are within 20% of one another at temperatures above 500 K, and the former lies between the two experimental measurements at ~ 298 K, being within approximately a factor of two of the experimental measurements. While the rate constants from Chavarrio Cañas [115] are 2.8

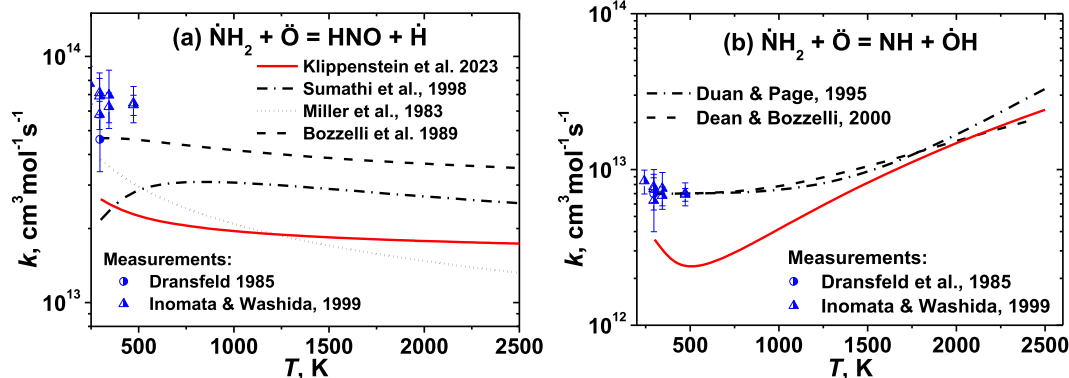


Fig. 5. Rate constants for $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}}$ to the products of (a) $\text{HNO} + \dot{\text{H}}$ [102,103,112] with experimental measurements from [99,104], and (b) $\dot{\text{N}}\text{H} + \dot{\text{O}}\text{H}$ [95,105] with experimental measurements from [99,104].

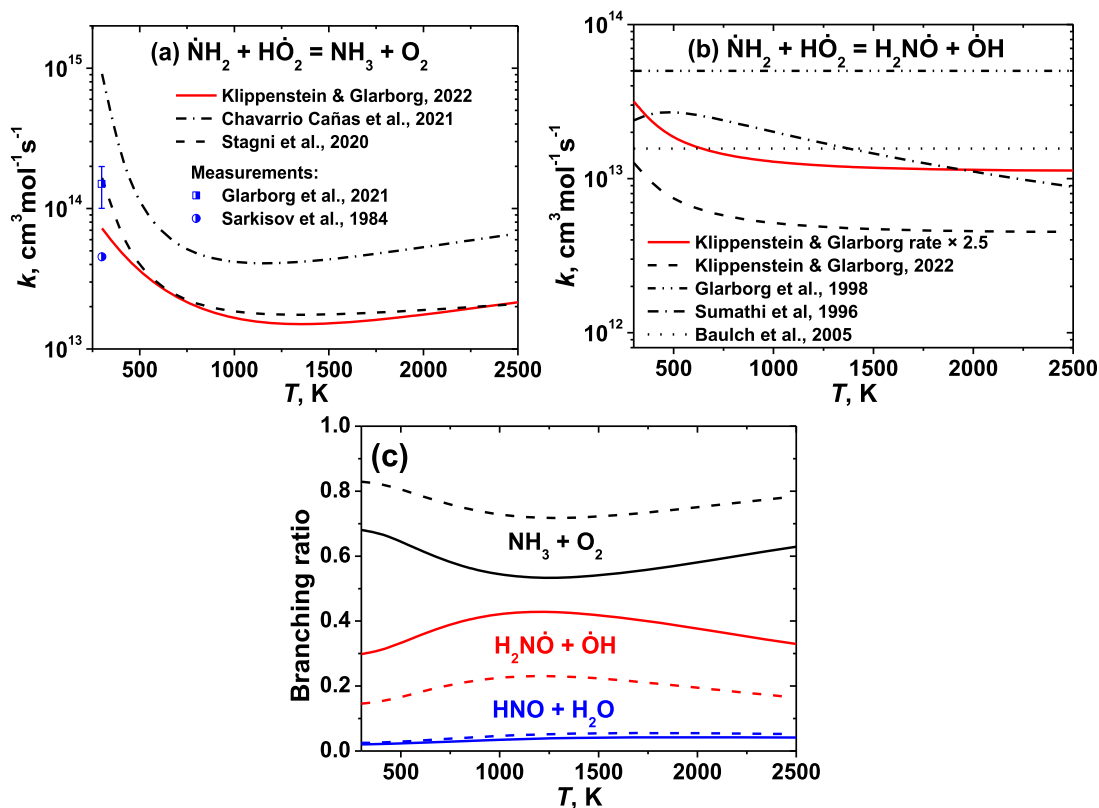
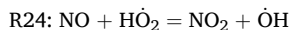


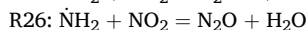
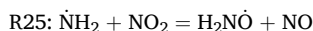
Fig. 6. Rate constants for $\dot{\text{N}}\text{H}_2 + \text{H}\ddot{\text{O}}_2$ to the products of (a) $\text{NH}_3 + \text{O}_2$ [36,114,115] with experimental measurements from [116,117] and (b) $\text{H}_2\dot{\text{N}}\text{O} + \dot{\text{O}}\text{H}$ [114, 118–120], and (c) their branching ratios employed in this work (solid lines) and from the originally reported Klippenstein and Glarborg [114] rate constants (dashed lines).

times faster than those calculated by Klippenstein and Glarborg [114] and Stagni [36], we use the calculated rate constants from Klippenstein and Glarborg [114] for R21 in the model. For R22, we found that the Klippenstein and Glarborg rate constants are the slowest among the rate constants reported [114,118–120], Fig. 6(b), and combining this rate constant with our adopted rates for R21 in the model leads to a significant under-estimation of fuel reactivity at $T < 1200$ K. Thus, we increased the Klippenstein and Glarborg rate constant for R22 by 2.5 times to achieve good agreement between our model predictions and the low-temperature (950–1200 K) experimental IDT data [33]. This adjustment results in an increase in the branching ratio of R22 increasing by approximately 18% in the temperature range 300–2000 K, Fig. 6(c).

With abundant $\text{H}\ddot{\text{O}}_2$ in the radical pool, the NO molecules in the mixture are oxidized into NO_2 more quickly by $\text{H}\ddot{\text{O}}_2$ via reaction R24:



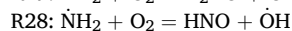
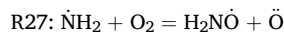
This converts relatively unreactive $\text{H}\ddot{\text{O}}_2$ radicals into more reactive $\dot{\text{O}}\text{H}$ radicals, promoting fuel reactivity at low temperatures. For $\text{NO} + \text{H}\ddot{\text{O}}_2 = \text{NO}_2 + \dot{\text{O}}\text{H}$ (R24), we employed a rate constants expression of $2.10 \times 10^{12} \exp(-497 \text{ cal}/RT) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ provided by Baulch et al. [120] that fits well the experimental measurements by Howard et al. [122] in the temperature range 232–2000 K. The resulting NO_2 molecules can react with $\dot{\text{N}}\text{H}_2$ radicals through the following barrierless addition reactions:



Again, the two product channels, R25 and R26, have opposite effects on fuel reactivity. R25 promotes reactivity because it converts NO_2 back

to NO which reacts favorably with $\text{H}\ddot{\text{O}}_2$ and $\dot{\text{N}}\text{H}_2$ (R8 and R24), further generating $\dot{\text{O}}\text{H}$ radicals; and R26 generates two stable molecules, thereby inhibiting reactivity. Klippenstein et al. [123] studied the kinetics of the reactions on the $\dot{\text{N}}\text{H}_2 + \text{NO}_2$ PES, with the rate constants for the two barrierless entrance channels determined using VRC-TST. Their total rate constants are in good agreement with available experimental data at temperatures below 1000 K [93,123,125,126], while the only measurements at $T > 1200$ K is from Song et al. [124], which is significantly faster, by approximately a factor of two, compared to Klippenstein et al. [123] and the total rate constant recommended by Glarborg et al. [45] lies between that measured by Song [124] and calculated by Klippenstein [123], Fig. 7(a). In this work, we adopt the rate constants calculated by Klippenstein et al. [123] for R25 and R26 with a modest adjustment (10%) in rate constant for R25, Fig. 7. This adjustment increases the branching ratio for R26 to 20% at 1000 K, but the total rate of $\dot{\text{N}}\text{H}_2 + \text{NO}_2$ still agrees with the experimental data measured in the temperature range 300–1000 K.

The reactions between $\dot{\text{N}}\text{H}_2$ and molecular oxygen O_2 are an important source of $\dot{\text{O}}\text{H}$ radicals at all temperatures. There are two competing product channels, but both promote reactivity:



R27 generates $\ddot{\text{O}}$ atoms which react with NH_3 to produce $\dot{\text{N}}\text{H}_2 + \dot{\text{O}}\text{H}$, while R28 produces $\dot{\text{O}}\text{H}$ radicals directly, hence both R27 and R28 promote reactivity. Klippenstein et al. [47] calculated both reactions through ab initio calculations employing VRC-TST. For R27, their rate constant is similar to the QRRK estimations from Bozzelli and Dean [102], nonetheless they are about 4.7 times slower than the experimentally derived rate constants from Henning et al. [127] in the temperature range 1450–2300 K, Fig. 8(a). In order to predict the

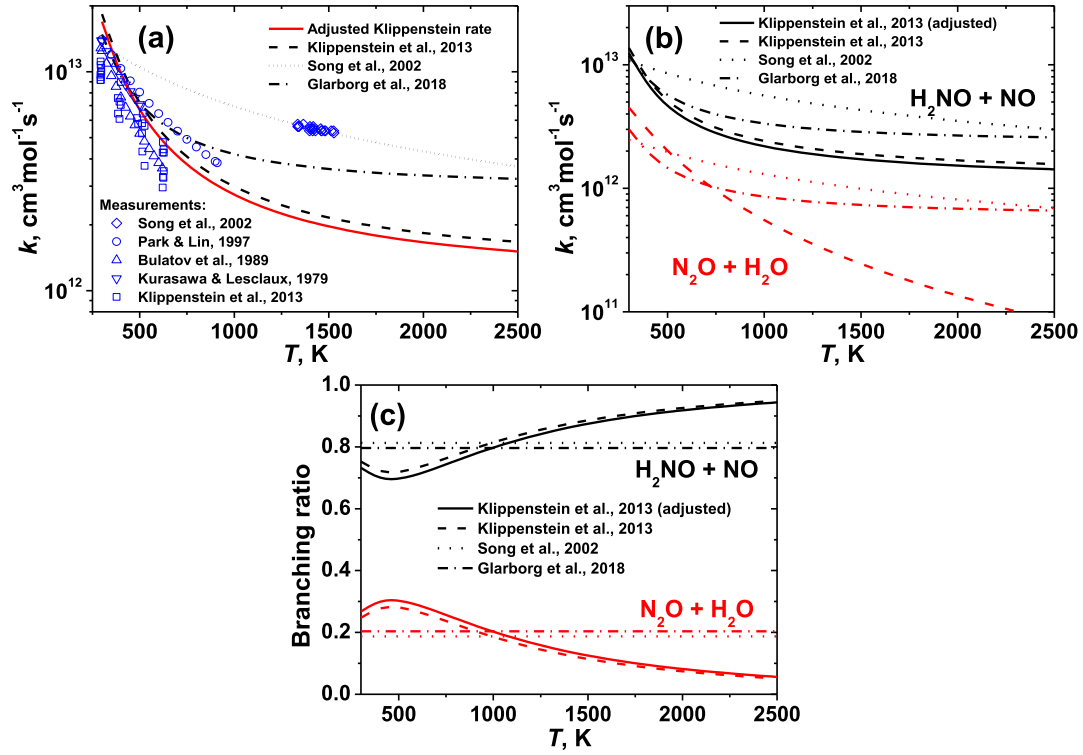


Fig. 7. (a) Individual rate constants and (b) branching ratios for the product channels of $\dot{\text{N}}\text{H}_2 + \text{NO}_2$, where solid lines represent the adjusted rate constants and the corresponding branching ratios used in this work; dashed lines, dotted lines and dashed-dotted lines represent those from Klippenstein et al. [123], Song et al. [124] and Glarborg et al. [45], respectively.

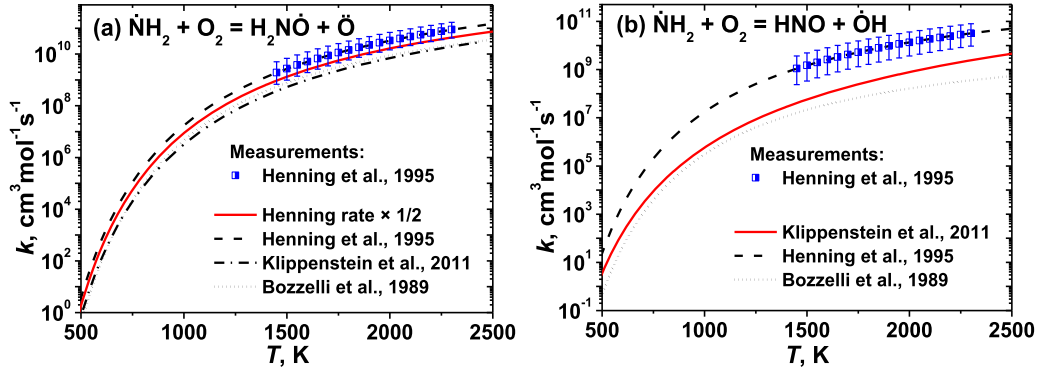
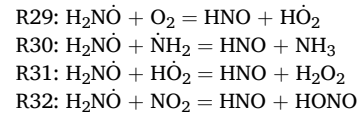


Fig. 8. Rate constants for $\dot{\text{N}}\text{H}_2 + \text{O}_2$ to the products of (a) $\text{H}_2\text{NO} + \dot{\text{O}}$ [47,102,127] with experimental measurements from [127], and (b) $\text{HNO} + \dot{\text{O}}\text{H}$ [47,102,127] with experimental measurements from [127].

low-temperature IDTs of pure NH_3 , we need to use a faster rate constant expression for R27 than that calculated by Klippenstein et al. Therefore, we employed the fitted rate constants expression from Henning et al. [127] for R27 with the A-factor decreased by a factor of two, which is within the error bars of their measured data points. Moreover, the adjusted Henning rate constants are within the estimated factor of two uncertainty in the Klippenstein et al. [47] calculations. For R28, the rate constants calculated by Klippenstein et al. [47] are 17.6 times slower than those measured by Henning et al. [127] in the temperature range 1450–2300 K, Fig. 8(b). The reported estimated uncertainty in the rate constant by Klippenstein et al. [47] is a factor of four at 1000 K. We employed the original rate constants for R28 reported by Klippenstein et al. [47] to avoid over-predictions in fuel reactivity and NO concentrations, since R28 produces $\dot{\text{O}}\text{H}$ and HNO directly, both of which are linked to NO formation.

H_2NO radicals are important intermediate species produced through the

following reactions: $\dot{\text{N}}\text{H}_2 + \dot{\text{H}}\text{O}_2$, $\dot{\text{N}}\text{H}_2 + \text{NO}_2$ and $\dot{\text{N}}\text{H}_2 + \text{O}_2$. H_2NO controls fuel reactivity at low temperatures, mainly because of its subsequent reactions with O_2 , $\dot{\text{N}}\text{H}_2$, $\dot{\text{H}}\text{O}_2$ and NO_2 .

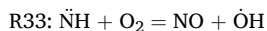


R29 promotes fuel reactivity, as it converts O_2 into $\dot{\text{H}}\text{O}_2$ radicals which are subsequently converted into $\dot{\text{O}}\text{H}$ radicals via R22 and R24 at low temperatures. Reactions R30–R32 are chain terminating and thus inhibit reactivity. Although the importance of reactions R29–R32 has been reported [36,45], accurate kinetic data for these four reactions is limited. Recently, Stagni and Cavallotti [128] performed high-level ab initio calculations on these reactions using the multi-reference CASPT2

method, where two outer transition states and one inner transition state for each individual reaction was considered in the calculations. The bottle-neck reactive fluxes for each reaction, which is largely determined by the inner transition state, was calculated using conventional transition state theory (CTST). The rate constants calculated by Stagni and Cavallotti [128] for R29–R32 are plotted in Fig. S8, and are compared to those reported in the literature. Song et al. studied the kinetics of R29 at the CCSD(T)/CBS//QCISD(T)/6–311G(d,p) level of theory [40], with the umbrella inversion motion of H_2NO radical treated as anharmonic vibration in the rate constants calculations. Glarborg et al. [45] adopted the Song rate constant for R29 in their model with the E_a adjusted by $-2.4 \text{ kcal mol}^{-1}$ in order to capture the flow reactor speciation data at fuel-lean conditions. It is interesting to note that the newly calculated rate constant for R29 by Stagni et al. is similar to the adjusted rate constants by Song by Glarborg et al. (within about 20% in the temperature range 500–2500 K), Fig. S8(a). The calculated rate constants for R29 from Stagni and Cavallotti [128] is used in our model. For R30–R32, however, the Stagni rate constants are not employed or are employed with a-factor-of-two adjustment. In our model, we have decreased the Dean and Bozzelli [95] rate constant by a factor of two for R30, increased the Stagni [128] rate constant by a factor of two for R31 and decreased the Glarborg [129] rate constant by a factor of two for R32, in order to achieve an overall best agreement with the low-temperature pure NH_3 IDTs experimental data from $\varphi = 0.5$ to 3.0 at $1050 \leq T \leq 1210 \text{ K}$ and 60 bar [33].

2.3. NO_x and N_2O chemistry

The formation and consumption of nitrogen oxides (NO_x) and nitrous oxide (N_2O) are coupled with the oxidation reaction pathways of fuel molecules and fuel-derived radicals that control fuel reactivity, Fig. 1. The NO_x reaction subset in our model is built largely based on the work of Sahu et al. [130] which is well-validated against C_1 – C_3 alkane/ NO_x experimental data [130–132]. The rate constants associated with thermal- NO_x reaction pathways, $\text{N}_2 + \ddot{\text{O}} = \text{N} + \text{NO}$, $\text{N} + \text{O}_2 = \text{NO} + \ddot{\text{O}}$ and $\text{N} + \text{OH} = \text{NO} + \dot{\text{H}}$ are updated here as shown in Fig. S9. At high temperatures NO is mainly produced by HNO unimolecular decomposition (R15) and the reaction of HNO with O_2 , $\dot{\text{H}}$ and $\dot{\text{N}}\text{H}_2$ (R16–R18); while at low-to-intermediate temperatures, NO formation from $\dot{\text{N}}\text{H}_2 + \text{NO}_2 = \text{H}_2\text{NO} + \text{NO}$ (R25) becomes favorable in addition to R15–R18. The upstream sources of HNO differ at different temperatures. At high temperatures HNO is mainly produced via $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}} = \text{HNO} + \dot{\text{H}}$ (R14), while at low-to-intermediate temperatures H-atom abstraction from H_2NO (R29–R31) are the major sources of HNO produced. Moreover, a considerable proportion of NO can be formed from NH reaction with O_2 :



Elishav et al. [78] reviewed rate constants and experimental measurements for $\dot{\text{N}}\text{H} + \text{O}_2 = \text{NO} + \dot{\text{O}}\text{H}$ (R33), and recommended the fitted rates reported by Römning [133], as they better aligned with both measured data at low temperatures in the literature and with their own experimental data measured in the temperature range 1000–2000 K. Thus, we use the Römning rate constant. The NO produced is mainly consumed through its reactions with $\dot{\text{N}}\text{H}_2$ (R8 and R9) and $\dot{\text{N}}\text{H}$.

At all temperatures NO mainly reacts with HO_2 producing NO_2 and $\dot{\text{O}}\text{H}$ (R24) which is one of the most important reactivity-promoting reactions at low-to-intermediate temperatures as discussed above. Through flux analyses discussed in Section 3, it is found that NO_2 reactions with $\dot{\text{N}}\text{H}_2$ (R25 and R26) and $\text{NO}_2 + \dot{\text{H}} = \text{NO} + \dot{\text{O}}\text{H}$ are the major pathways consuming NO_2 . N_2O is mainly produced from the reaction of $\dot{\text{N}}\text{H}$ with NO at all temperatures:



Moreover, at low temperatures, N_2O is formed via $\dot{\text{N}}\text{H}_2 + \text{NO}_2 = \text{N}_2\text{O} + \text{H}_2\text{O}$ (R26). Available rate constant expressions for $\dot{\text{N}}\text{H} + \text{NO} = \text{N}_2\text{O} + \dot{\text{H}}$ (R34) in the literature [47,109,120,134] are illustrated in Fig. S10. These data vary by about a factor of three, with the rate constant recommended by Baulch et al. [120] being intermediate to the others. In this work, we adopted the Baulch rate constant for R34, but we have decreased it by 20% to better capture both the time-resolved and temperature-resolved experimentally measured N_2O concentration profiles. Through flux analyses it is found that $\text{N}_2\text{O} + \dot{\text{H}} = \text{N}_2 + \dot{\text{O}}\text{H}$ is the dominant N_2O consumption pathway. For this reaction the rate constant expression from Marshall et al. [135] is used in our model, which provides a good fit to the available kinetics measurements in temperature range 300–2500 K, Fig. S11. Recently, Meng et al. [136] found that in their ab initio reaction kinetics study it is possible for $\text{N}_2\text{O} + \dot{\text{H}}$ to be stabilized into $\text{HNN}\dot{\text{O}}$ isomers, and proposed a reaction sub-mechanism of $\text{HNN}\dot{\text{O}}$ isomers oxidation. We incorporated their proposed $\text{HNN}\dot{\text{O}}$ sub-mechanism in our model in order to better describe the NO_x chemistry.

3. Model validation and analysis

The NH_3 and NH_3/H_2 combustion simulations were performed using Cantera [137]. Fuel ignition in shock tubes were simulated assuming a constant-volume, adiabatic process, with the definitions of IDT consistent with those in the literature experiments; fuel IDTs in RCMs were simulated by importing the volume profiles derived from non-reactive pressure traces, provided in the literature experiments, to account for facility effects. In one-dimensional flame simulations, the multi-component method was employed to calculate species transport properties and thermal diffusion mass fluxes were considered. Tight criteria for grid refinements were employed to ensure sufficient precision, with the spacing ratio on either side of each grid point set to 2.0, and maximum differences in the computed values and the slopes between two adjacent grid points both set to 0.01. For burner-stabilized flame structure simulations, experimentally measured temperature profiles were employed. JSRs were simulated as an open system with constant pressure and temperature, while PFRs were simulated as either a one-dimensional constant-pressure reacting flow employing experimentally measured temperature profiles or a zero-dimensional constant-pressure and temperature batch reactor employing temperature-dependent residence times.

3.1. NH_3/H_2 high-temperature IDT validation

The model predicted high-temperature IDTs of pure NH_3 and NH_3/H_2 mixtures are compared to the experimental shock tube data of Mathieu et al. [12], Chen et al. [13] and Baker et al. [30] in Figs. 9, S12–S14 respectively. Our model captures the pure NH_3 high-temperature IDTs at all equivalence ratios of 0.5, 1.0, 2.0 and pressures of 1.2–28.9 atm quite well; the predicted NH_3/H_2 high-temperature IDTs are also in good agreement with the experimental data for mixtures with various H_2 fractions ranging from 5 to 70% at $\varphi = 1.0$ and at pressures in the range 1.2–10 atm. Brute-force IDT sensitivity analyses and reaction path flux analyses for some selected conditions are shown in Figs. 10, S15 and S16, respectively.

Fig. 10 shows the sensitivity analyses to IDTs for stoichiometric NH_3/H_2 mixtures with H_2 ratios varying from 0%, 5% and 30% at 1800 K and 10 atm, and the corresponding reaction path fluxes for the mixtures with 0% and 30% H_2 . For NH_3/H_2 fuel mixtures, H_2 reactivity is initiated by H-atom abstraction from H_2 by $\ddot{\text{O}}$ and $\dot{\text{O}}\text{H}$, these being the major source of $\dot{\text{H}}$ atoms in the radical pool and significantly increases the fuel reactivity at all temperatures. While for pure NH_3 , $\dot{\text{H}}$ atoms are mainly supplied by NNH decomposition, $\text{NNH} = \text{N}_2 + \dot{\text{H}}$, and $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H} = \text{N}_2\text{H}_2 + \dot{\text{H}}$ (R10). The reactivity inhibiting effect of $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}} = \text{HNO} + \dot{\text{H}}$ (R14) increases for NH_3/H_2 mixtures because it competes with the chain-

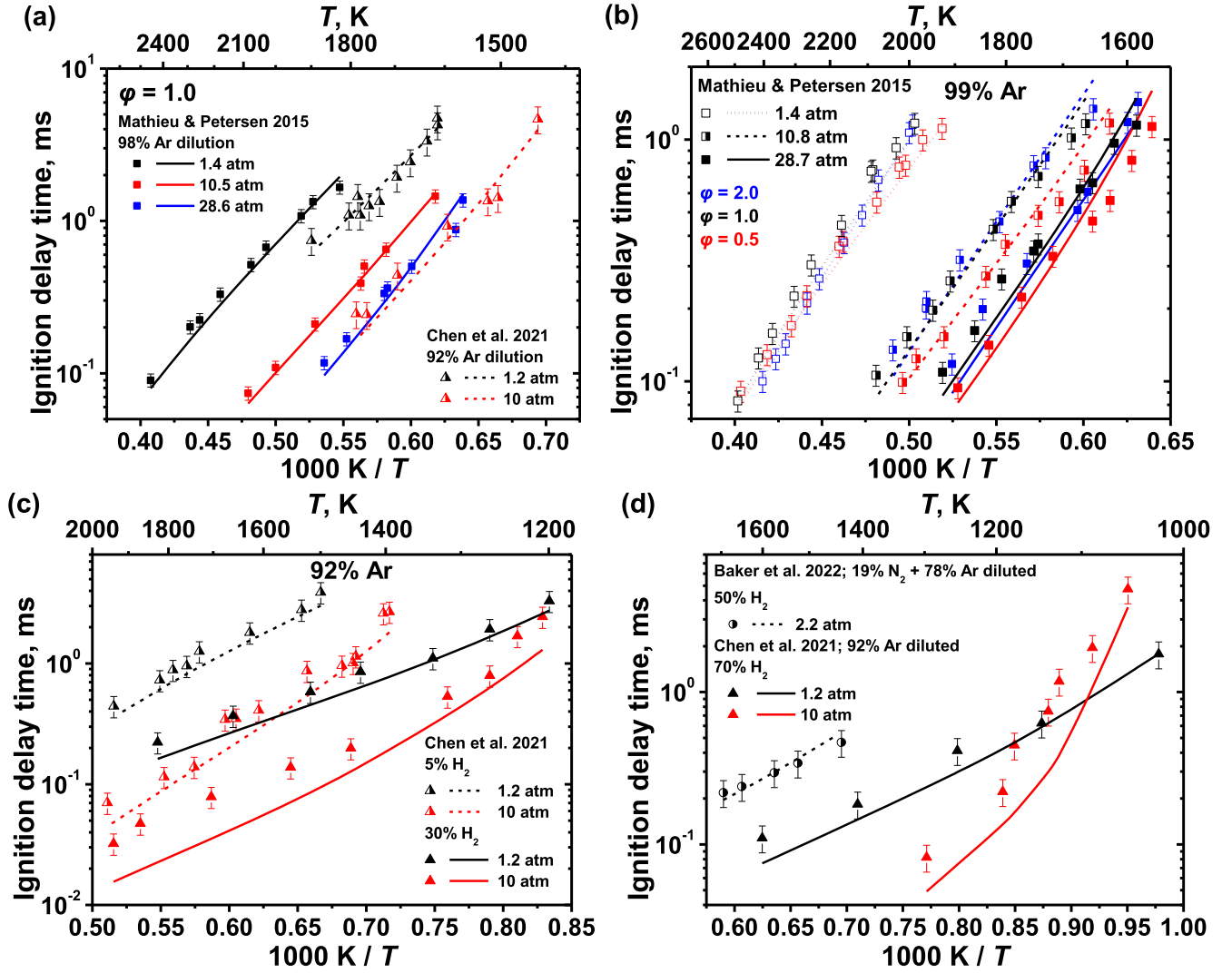


Fig. 9. IDTs obtained from shock tubes for pure NH_3 at (a) $\phi = 1.0$ and $p = 1.4$ – 28.6 atm, (b) $\phi = 0.5, 2.0$ and $p = 1.4$ – 28.9 atm; for 95/5 and 70/30 NH_3/H_2 mixtures at (c) $\phi = 1.0$, $p = 1.2, 10$ atm, and 50/50 and 30/70 NH_3/H_2 mixtures at (d) $\phi = 1.0$, $p = 1.2$ – 10 atm. Symbols are experimental data from Mathieu and Petersen [12], Chen et al. [13] and Baker et al. [30], while lines are predictions of the model developed in this work.

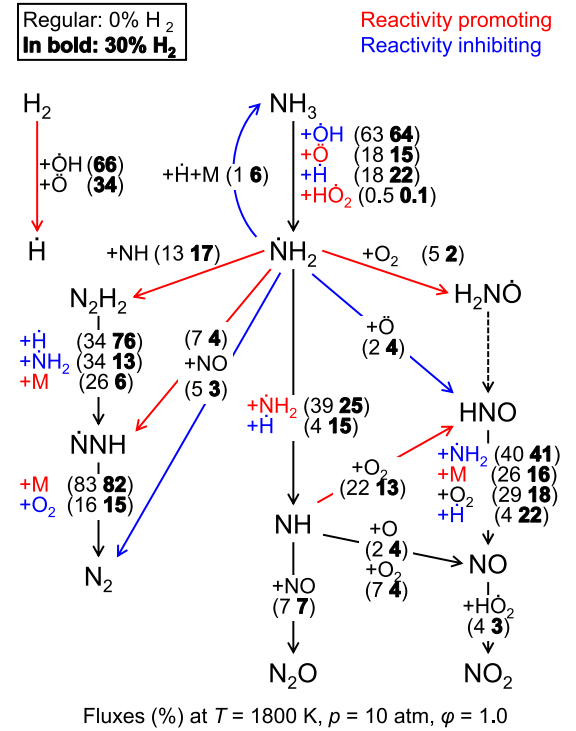
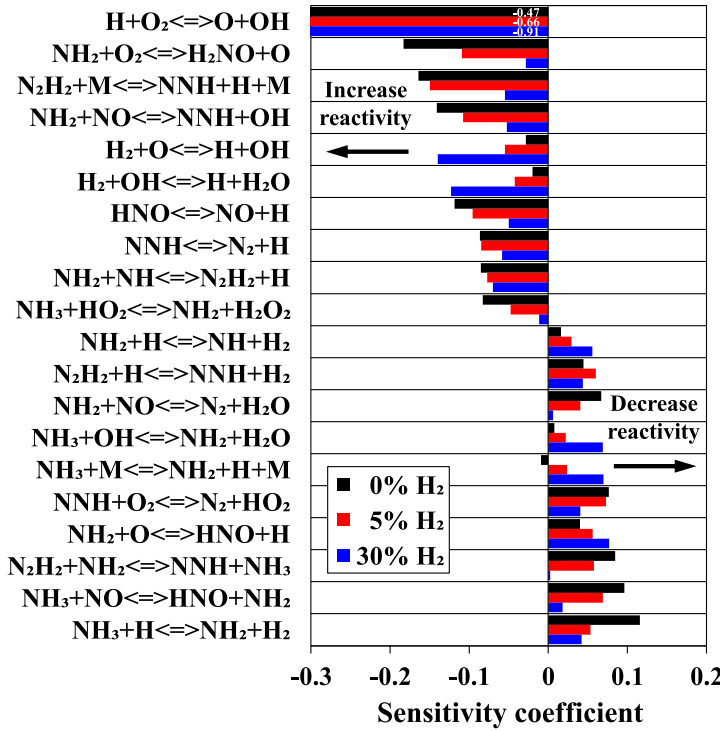


Fig. 10. Brute-force IDT (time at maximum $\dot{\text{O}}\text{H}$ concentration) sensitivity analyses for 100/0, 95/5 and 70/30 NH_3/H_2 mixtures at $\phi = 1.0$, $T = 1800\text{ K}$, $p = 10\text{ atm}$, and the corresponding flux analyses for NH_3 consumption for 100/0 and 70/30 NH_3/H_2 mixtures.

branching reaction $\text{H}_2 + \ddot{\text{O}} = \dot{\text{H}} + \dot{\text{O}}\text{H}$. Moreover, H-atom abstraction from NH_3 by $\dot{\text{O}}\text{H}$ (R4) competes with $\text{H}_2 + \dot{\text{O}}\text{H} = \dot{\text{H}} + \text{H}_2\text{O}$ and hence becomes more inhibiting for NH_3/H_2 mixtures. The concentration of $\dot{\text{H}}$ atoms in the radical pool of NH_3/H_2 mixtures is higher than for pure NH_3 , hence more $\dot{\text{N}}\text{H}_2$ radicals undergo reaction with $\dot{\text{H}}$ atoms (R-1) and lead to chain termination; more N_2H_2 and HNO molecules react with $\dot{\text{H}}$ atoms (R12 and R17) than directly decompose. The important reactions producing NO , N_2O and NO_2 at high-temperature conditions in the NH_3/H_2 system are also shown in the reaction path flux diagram in Fig. 10. NO is mainly produced from HNO decomposition and abstraction reactions (R15–18), and $\dot{\text{N}}\text{H}$ addition reactions with O_2 (R33) and $\ddot{\text{O}}$ atoms. The addition of H_2 to NH_3 fuel does not change these formation pathways but tends to enhance NO formation because it produces higher concentrations of HNO and $\dot{\text{N}}\text{H}$. At high temperatures N_2O is almost completely produced by $\dot{\text{N}}\text{H} + \text{NO} = \text{N}_2\text{O} + \dot{\text{H}}$ (R34) while NO_2 is almost entirely produced from $\text{NO} + \text{H}\ddot{\text{O}}_2 = \text{NO}_2 + \dot{\text{O}}\text{H}$ (R24). Due to the scarcity of $\text{H}\ddot{\text{O}}_2$ radicals at high temperatures, only a relatively small concentration of NO_2 is formed.

3.2. NH_3/H_2 laminar flame speeds validation

Experimentally measured literature laminar flame speeds for NH_3/H_2 mixtures at different H_2 fractions, equivalence ratios (ϕ), unburnt gas temperatures (T_i) and pressures, together with our model predictions, are shown in Fig. 11. While representative flame speed validation results are shown in the main text, those for all targets listed in Table 2 is available as Figs. S17–S25 in the SM. In general, our model captures the NH_3/H_2 flame speeds at most conditions very well. Fig. 11(a) shows the dependence of pure NH_3 flame speeds on T_i . Although the unburnt fuel-air mixtures of pure NH_3 were heated to 200 °C, their flame speeds are slower than 20 cm s^{-1} and not compete with those of methane (25–35 cm s^{-1}) at room temperature. The Han et al. data [18,20] are slower than those reported in [19,21,24] by $\sim 2\text{ cm s}^{-1}$ at the same conditions on the fuel-rich side. The model predicted pure NH_3 flame speeds are in good agreement with the latter [19,21,24] at 298–423 K but are slightly faster

at 473 K. As illustrated in Figs. 11(b) and 11(c), NH_3/H_2 mixtures with 30% or more H_2 start to have comparable reactivity to pure methane. Our model captures the dependence of NH_3/H_2 on H_2 ratio and T_i quite well, especially for the mixtures with 10–50% H_2 . Moreover, our model predicts the NH_3/H_2 flame speeds at $\phi < 1.2$ and at different pressures up to 10 atm reasonably well, but it slightly over-predicts the fuel-rich experimental data, Figs. 11(d) and S21. Fig. 11(d) also shows that there is considerable scatter in available experimental data for the NH_3/H_2 (40%) mixture at atmospheric condition on the fuel-rich side, with the data from Lhuillier et al. [21] and Zitouni et al. [28] $\sim 3\text{ cm s}^{-1}$ higher than those from Han et al. [20], Wang et al. [22] and Gotama et al. [27]. Since most NH_3/H_2 flame speed experiments were conducted at atmospheric pressure, higher pressure data would be useful to test model predictions.

Sensitivity and reaction path flux analyses to flame speed at some selected conditions are shown in Figs. 12, S26 and S27, respectively. We observe that fuel reactivity is highly sensitive to reactions producing $\dot{\text{H}}$ atoms which trigger the important $\dot{\text{H}} + \text{O}_2 = \ddot{\text{O}} + \dot{\text{O}}\text{H}$ chain branching reaction. Fig. 12 shows the important reactions for the NH_3/H_2 (30%) mixture at $\phi = 0.8, 1.1$ and 1.4, respectively. $\dot{\text{H}}$ atoms are mainly produced from H_2 initiation reactions with $\dot{\text{O}}\text{H}$ and $\ddot{\text{O}}$, and NH_3 oxidation reactions $\dot{\text{N}}\text{NH} = \text{N}_2 + \dot{\text{H}}$, $\dot{\text{N}}\text{H}_2 + \dot{\text{N}}\text{H} = \text{N}_2\text{H}_2 + \dot{\text{H}}$ (R10) and $\dot{\text{N}}\text{H}_2 + \ddot{\text{O}} = \text{HNO} + \dot{\text{H}}$ (R14). R14 competes with two chain-branching reactions ($\text{NH}_3 + \ddot{\text{O}} = \dot{\text{N}}\text{H}_2 + \dot{\text{H}}$ (R3) and $\text{H}_2 + \ddot{\text{O}} = \dot{\text{H}} + \dot{\text{O}}\text{H}$), hence it is highly inhibiting. R10 serves as an important source of $\dot{\text{H}}$ atoms, with the product N_2H_2 molecules further producing $\dot{\text{N}}\text{NH}$ radicals. On the other hand, the reactions $\dot{\text{N}}\text{H}_2 + \text{N} \rightarrow \text{N}_2 + \dot{\text{H}} + \dot{\text{H}}$ and $\dot{\text{N}}\text{H} + \dot{\text{N}}\text{H} \rightarrow \text{N}_2 + \dot{\text{H}} + \dot{\text{H}}$ produce $\dot{\text{H}}$ atoms, and thus promote reactivity. H-atom abstraction by $\dot{\text{H}}$ atoms from NH_3 (R2), $\dot{\text{N}}\text{H}_2$ (R5) and HNO (R17), or $\dot{\text{H}}$ addition to $\dot{\text{N}}\text{H}_2$ (R-1) all consume $\dot{\text{H}}$ atoms and thus inhibit reactivity. Moreover, $\dot{\text{N}}\text{H}_2$ reactions with NO (R8 and R9) are also important, particularly at fuel-lean conditions where NO is abundant. Note that based on the 0 K formation enthalpies in the ATcT database [53] and the calculated [N_2H_2] PESs by Marshall et al. [138], the exothermicity of H_2NN formation from

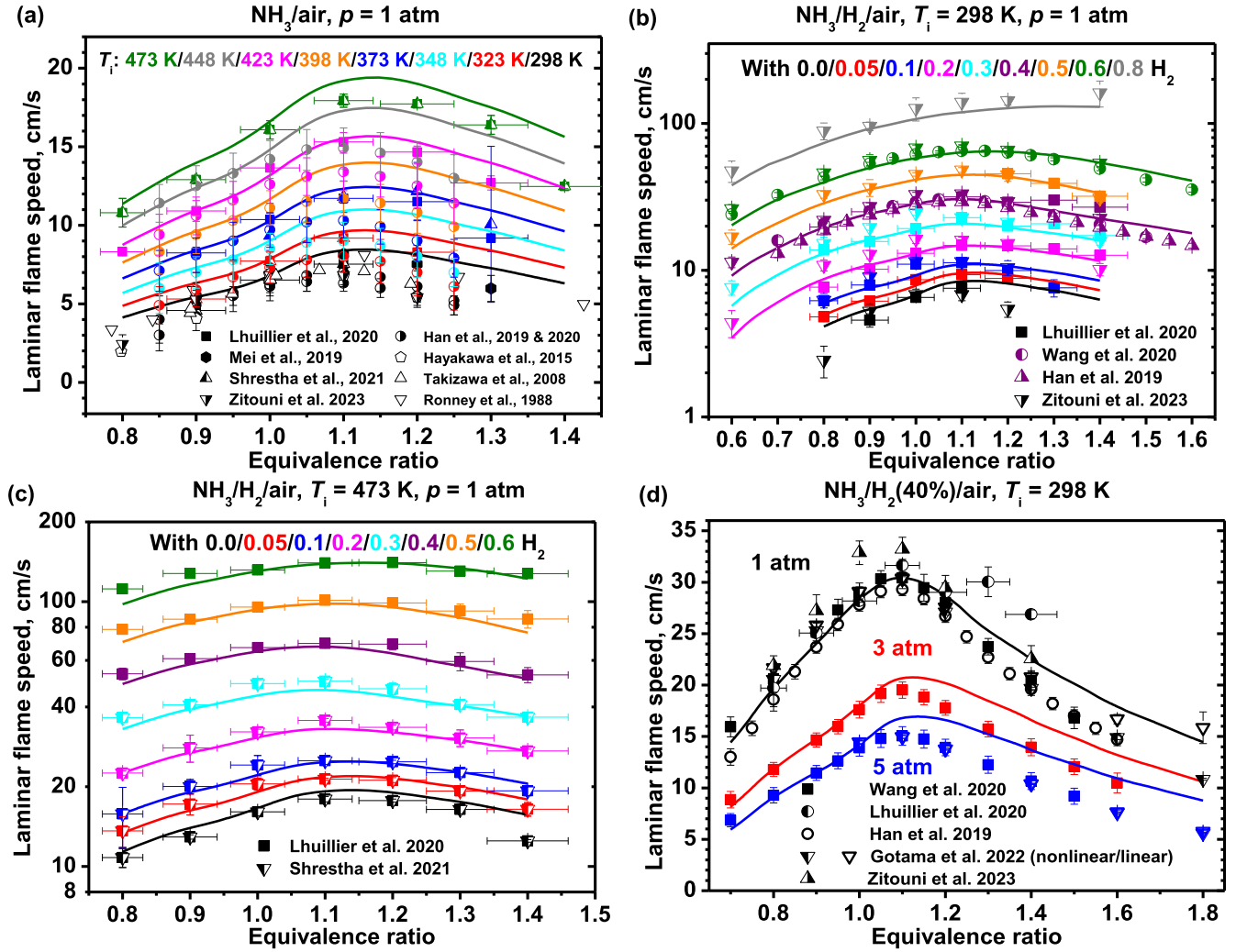


Fig. 11. (a) Pure NH_3 flame speeds at 1 atm and different T_i ; NH_3/H_2 (100/0 to 20/80) flame speeds at 1 atm and T_i of (b) 298 K, (c) 473 K; and (d) NH_3/H_2 (60/40) flame speeds at 298 K, and pressures of 1–5 atm. Symbols are experimental data from [17–22,24,27,28,140,141], while lines are predictions of the model developed in this work.

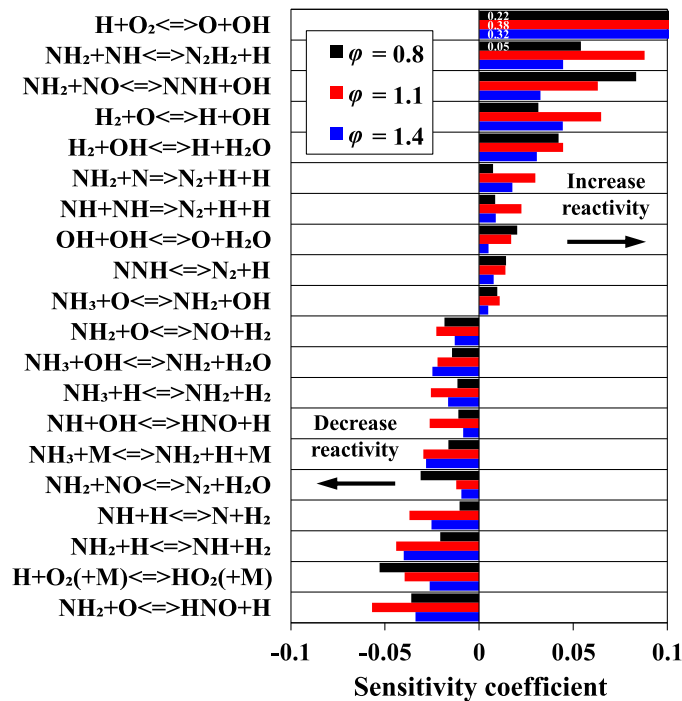


Fig. 12. Flame speed sensitivity analyses for 70/30 NH_3/H_2 mixtures in air at $T_1 = 373 \text{ K}$, $p = 1 \text{ atm}$ and $\phi = 0.8\text{--}1.4$.

$\dot{\text{N}}\text{H}_2 + \text{N}$ is sufficient for H_2NN to undergo the subsequent elimination reaction forming $\text{N}_2 + \text{H}_2$, with its transition state staying about 33 kcal mol^{-1} lower than the energy of $\dot{\text{N}}\text{H}_2 + \text{N}$. We included $\dot{\text{N}}\text{H}_2 + \text{N} = \text{N}_2 + \text{H}_2$ in our model, and have estimated the ratio of $\text{N}_2 + \dot{\text{H}} + \dot{\text{H}}$ and $\text{N}_2 + \text{H}_2$ formation reactions to be 65:35 by employing a total rate constant of $7 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ for $\dot{\text{N}}\text{H}_2 + \text{N}$ based on the kinetic measurements of Dransfeld and Wagner at 298 K [139]. Further study on the kinetics of $\dot{\text{N}}\text{H}_2 + \text{N}$ reactions is recommended.

3.3. Low-to-intermediate temperature NH_3/H_2 IDT validation

The NH_3/H_2 low-temperature IDTs measured in an RCM and intermediate-temperature IDTs measured in shock tube from the literature, along with our model predictions, are shown in Figs. 13, S28–S35. Our model predictions are in good agreement with the pure NH_3 experimental data from Dai et al. [33] and Shu et al. [29], but show some discrepancies with the data from He et al. [31]. The black squares and triangle in Fig. 13(a) were taken at the same pressure and mixture composition from Dai et al. [33] and He et al. [31], respectively. They are in reasonable agreement with each other at the overlapping temperature range. However, the simulated results obtained by using the corresponding volume histories from the two works diverge, which indicates that there may be uncertainties in the non-reactive pressure traces obtained from the experiments. Figs. 13(b) and (c) show the NH_3/H_2 low-temperature IDTs literature measurements at $\phi = 0.5$ and 1.0, pressures of 20–60 bar with H_2 fractions of 5–25% compared with the model predictions. Again, our model captures the experimental data from Dai et al. [33] quite well and predicts both the first-stage and total IDTs for NH_3/H_2 (25%) measured by Liao et al. [34] reasonably well, but it is not able to capture the measured data from He et al. [31]. The half-filled squares and triangles in Fig. 13(b) are experimental data for NH_3/H_2 (5%) taken at the same pressures and mixture compositions

from Dai et al. [33] and He et al. [31], respectively. At 20 bar, the two sets of measured data deviate by about a factor of three at overlapping temperatures while at 40 bar the data agree well with each other. By employing the corresponding volume profiles in simulations, our model predictions stay in line with the measurements from Dai et al. [33] but faster than the data from He et al. [31]. Moreover, our model predicts the species concentration time histories measured by Liao et al. [34] in RCM very well, Fig. S35.

Brute-force IDT sensitivity analyses for pure NH_3 at 1150 K, 60 atm and $\phi = 0.5\text{--}2.0$, NH_3/H_2 at 1050 K, 40 atm and $\phi = 0.5$ with H_2 ratios of 5–20%, and the corresponding flux analyses are shown in Figs. 14 and S36. At low temperatures, $\text{H}\dot{\text{O}}_2$ radicals are abundant in the radical pool and $\text{H}\dot{\text{O}}_2$ addition reactions to $\dot{\text{N}}\text{H}_2$ (R21–23) become favorable. Fuel reactivity is highly sensitive to the branching ratio of formations of $\text{NH}_3 + \text{O}_2$ (R21) and $\text{H}_2\text{N}\dot{\text{O}} + \dot{\text{O}}\text{H}$ (R22) from $\dot{\text{N}}\text{H}_2 + \text{H}\dot{\text{O}}_2$, because R21 leads to chain termination while R22 produce reactive $\dot{\text{O}}\text{H}$ radical and $\text{H}_2\text{N}\dot{\text{O}}$ radical which are very important intermediates at low-temperature combustion regime. $\text{H}_2\text{N}\dot{\text{O}}$ radicals are mainly produced from $\text{NH}_2 + \text{NO}_2 = \text{H}_2\text{N}\dot{\text{O}} + \text{NO}$ (R25) and $\dot{\text{N}}\text{H}_2 + \text{O}_2 = \text{H}_2\text{N}\dot{\text{O}} + \dot{\text{O}}$ (R27) in addition to R22, which are all important reactivity-promoting reaction at low temperatures. The $\text{H}_2\text{N}\dot{\text{O}}$ radicals then undergo H-atom abstraction reactions by O_2 , $\dot{\text{N}}\text{H}_2$, $\text{H}\dot{\text{O}}_2$ and NO_2 (R29–32) to form HNO molecules, which are also significant. While R29 is the only chain-propagation reaction among the four reaction pathways, it promotes fuel reactivity and competes with the other three chain-termination reactions of R30–R32. The HNO molecules are the major source of NO in the system. After they were formed through $\text{HNO} + \text{O}_2 = \text{NO} + \text{H}\dot{\text{O}}_2$, the NO molecules are oxidized by $\text{H}\dot{\text{O}}_2$ into NO_2 (R24) or undergo addition reactions to $\dot{\text{N}}\text{H}_2$ (R8 and R9), dictating fuel reactivity and supplying NO_2 for R25. As the H_2 fraction in the NH_3/H_2 mixtures increases from 5% to 20%, the most important reactivity-promoting reactions shift from $\dot{\text{N}}\text{H}_2 + \text{NO}_2 = \text{H}_2\text{N}\dot{\text{O}} + \text{NO}$ (R25) and $\dot{\text{N}}\text{H}_2 + \text{H}\dot{\text{O}}_2 = \text{H}_2\text{N}\dot{\text{O}} + \dot{\text{O}}\text{H}$ (R22) to $\dot{\text{H}} + \text{O}_2 = \dot{\text{O}} + \dot{\text{O}}\text{H}$ and H-atom abstraction from H_2 by $\dot{\text{N}}\text{H}_2$ (R–2), Fig. 14(b).

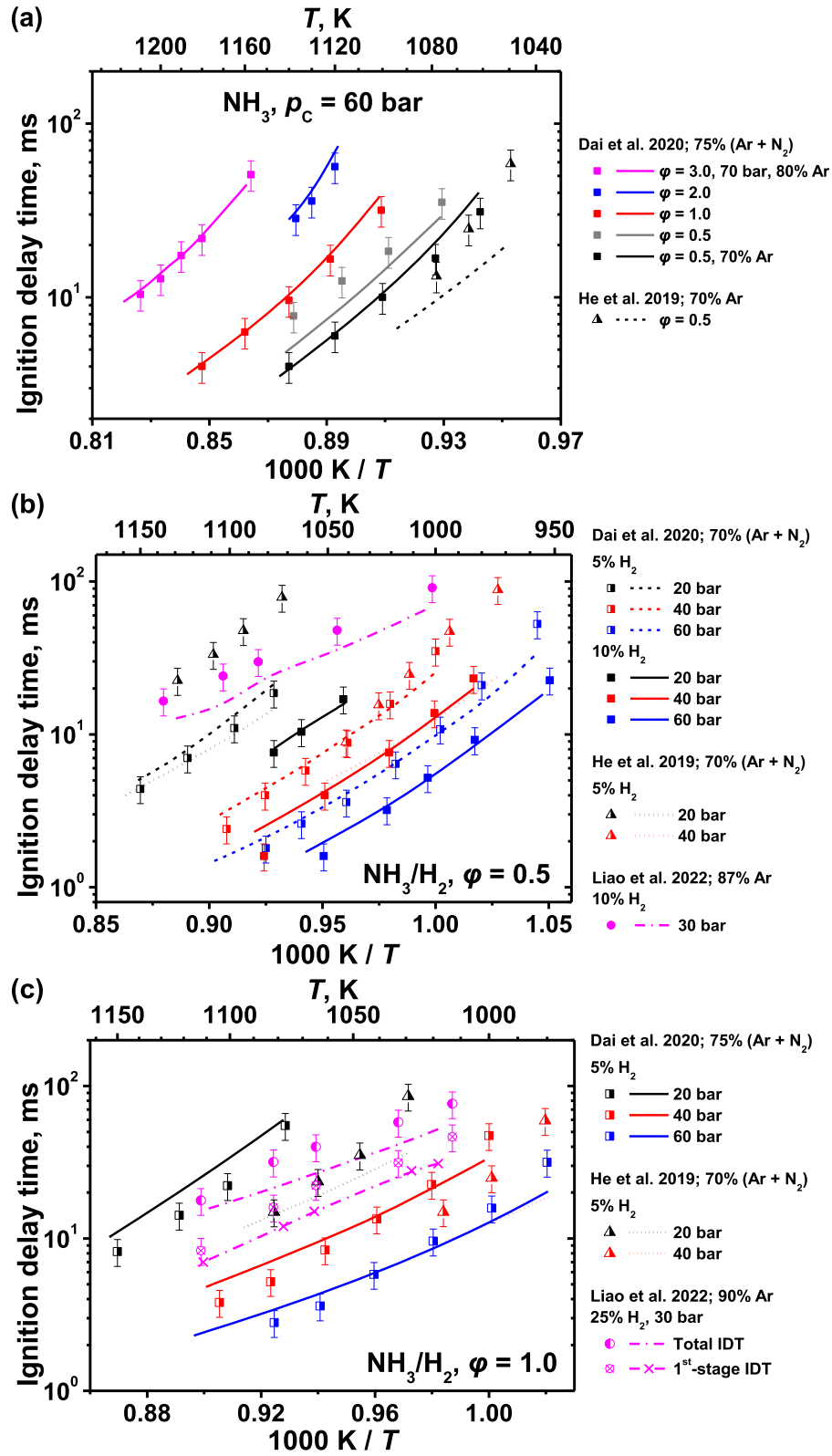


Fig. 13. RCM IDTs for (a) pure NH_3 at 60 bar, $\phi = 0.5$ –3.0; for 95/5, 90/10 and 75/25 NH_3/H_2 mixtures at (b) $\phi = 0.5$, 20–60 bar, and (c) $\phi = 1.0$, 20–60 bar. Symbols are experimental data from [29,31,33,34], while lines are predictions of the model developed in this work.

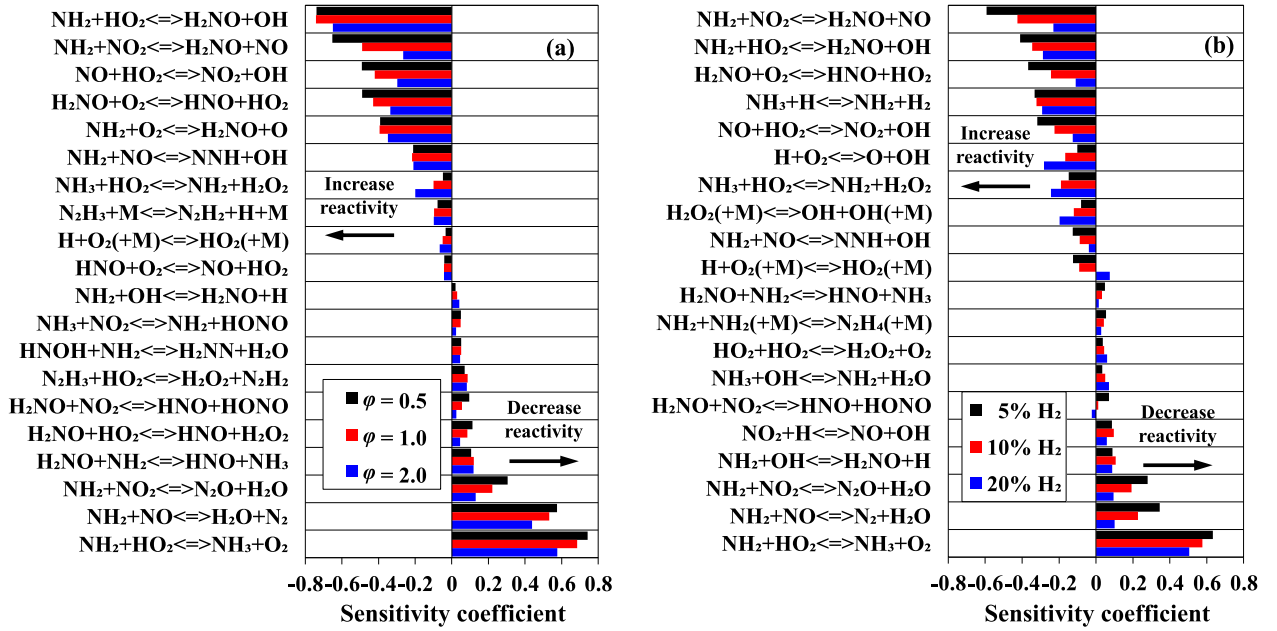


Fig. 14. Brute-force IDT (time at maximum dp/dt) sensitivity analyses for (a) pure NH_3 at $\phi = 0.5\text{--}2.0$, $T = 1150\text{ K}$, $p = 60\text{ atm}$, and (b) 95/5, 90/10 and 80/20 NH_3/H_2 mixtures at $\phi = 0.5$, $T = 1050\text{ K}$, $p = 40\text{ atm}$, H_2 fractions of 5–20 %.

3.4. Speciation during NH_3/H_2 pyrolysis and oxidation

The model developed in this work gives satisfactory predictions for most speciation experimental literature data summarized in Table 3, as illustrated in Figs. 15–17, S37–S57. Alturaifi et al. recently probed the NH_3 consumption [15] and N_2O formation [16] processes in a shock

tube for NH_3/H_2 pyrolysis and pure NH_3 oxidation, respectively, at high temperature and near atmospheric pressure conditions. As shown in Figs. 15, S44 and S45, our model predicts well NH_3 concentration time history for pure NH_3 and NH_3/H_2 pyrolysis compared to the data measured by Alturaifi et al. Without the presence of oxidizer, the NH_3 consumption is slowed by blending H_2 mainly because the increased

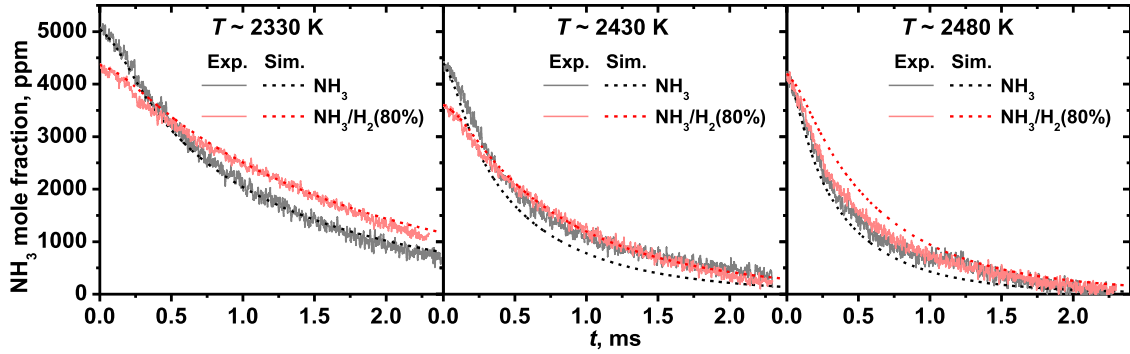


Fig. 15. NH_3 concentration time histories during pure NH_3 and 20/80 NH_3/H_2 mixtures pyrolysis in Ar in a shock tube at $p = 1.2\text{ atm}$, with experimental data from Alturaifi et al. [15].

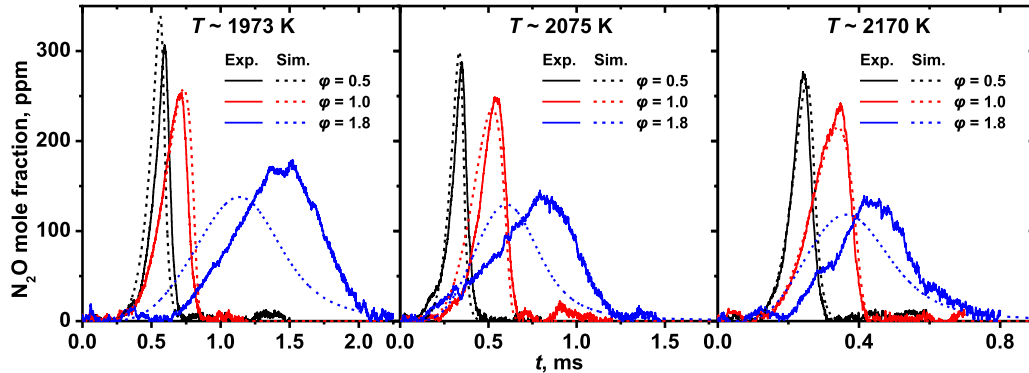


Fig. 16. N_2O concentration time histories during pure NH_3 oxidation in Ar in a shock tube at $p = 1.2\text{ atm}$, $\phi = 0.5\text{--}1.8$, with experimental data from Alturaifi et al. [16].

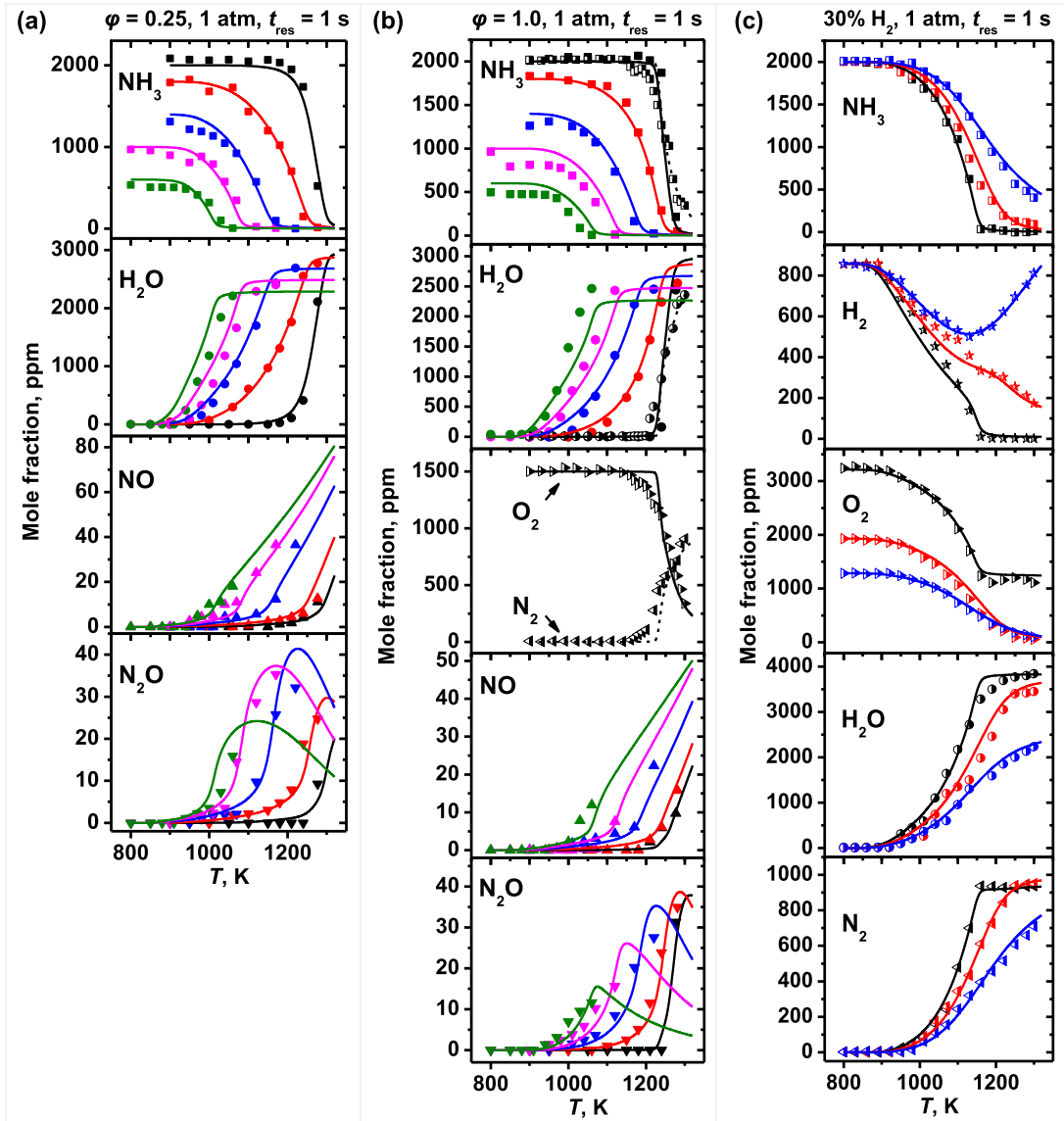


Fig. 17. NH_3/H_2 speciation in JSR at different conditions. (a) NH_3 with 0 %, 10 %, 30 %, 50 % and 70 % H_2 at $\phi = 0.25$ (in black, red, blue, magenta and green, respectively). (b) NH_3 with 0 %, 10 %, 30 %, 50 % and 70 % H_2 at $\phi = 1.0$ (in black, red, blue, magenta and green, respectively). (c) NH_3/H_2 (30 %) at $\phi = 0.6, 1.0$ and 1.5 (in black, red and blue, respectively). Solid symbols are experimental data from Zhang et al. [37] while half-filled symbols are experimental data from Osipova et al. [39]. Lines are predictions of the model developed in this work.

concentrations of $\dot{\text{H}}$ atoms produced from H_2 suppress the unimolecular decomposition of NH_3 (R1). Fig. 16 shows the dependence of N_2O concentration time histories on equivalence ratio during NH_3 oxidation. At high temperatures and at near atmospheric pressure, the peak concentration of N_2O is higher and occurs more quickly at fuel-lean compared to fuel-rich conditions. Our model captures the shapes of the measured N_2O time histories at $\phi = 0.5$ and 1.0 quite well, but slightly under-predicts the peak N_2O concentrations at $\phi = 1.8$ and over-predicts their time of production, Figs. 16 and S46. As demonstrated by the reaction path flux analyses, Figs. 10 and S27, N_2O produced in both pure NH_3 and NH_3/H_2 mixtures mainly comes from the reaction of $\dot{\text{N}}\text{H}$ with NO (R34) at high-temperature conditions.

The speciation data measured by Zhang et al. [37] and Osipova et al. [39] respectively in JSRs for NH_3/H_2 mixtures, at the same pressure of 1 atm and at a residence time (t_{res}) of 1 s, at different equivalence ratios of 0.25–1.5 and H_2 fractions of 0–70%, are presented together with our model predictions in Fig. 17. The model captures the NH_3 , H_2O , NO and N_2O concentrations at $\phi = 0.25$ and different H_2 ratios reasonably well for the fuel-lean data, although it slightly over-predicts fuel reactivity for

H_2 ratios higher than 50%, Fig. 17(a). The model predictions for NH_3 , H_2 , O_2 , H_2O and N_2 concentrations for NH_3/H_2 (30%) at $\phi = 0.6$ are also in excellent agreement with the experimental data, Fig. 17(c). At stoichiometric and fuel-rich ($\phi = 1.5$) conditions, the model again accurately predicts the concentrations of NO and N_2O in addition to those of the main species, NH_3 , H_2 , O_2 , H_2O and N_2 for mixtures with H_2 fractions ranging from 0% to 70%, Figs. 17(b) and 17(c). As discussed earlier, at low temperatures NO is mainly produced from the oxidation of $\text{H}_2\text{N}\dot{\text{O}}$ radicals. Meanwhile N_2O is mainly produced by the reaction of $\dot{\text{N}}\text{H}_2$ with NO_2 .

4. Conclusions

In this study, the combustion chemistry of pure NH_3 and NH_3/H_2 mixtures over a wide range of engine-relevant conditions has been investigated. A detailed NH_3/H_2 kinetic model has been developed through an extensive evaluation of available kinetic literature data for the important reaction pathways and species thermochemistry parameters. This has been comprehensively validated against available

experimental IDTs, flame speeds and speciation data in the literature. In general, our model can capture most of the validation targets within 20%. Girhe et al. [142] recently performed quantitative evaluations of the performances of 16 different NH_3/H_2 combustion kinetic models against 35 literature experimental data sets for NH_3/H_2 , using both the Error Function and Curve Matching (CM) approaches. The fidelity of our model is also demonstrated therein [142] where the overall score of our model, evaluated using the CM approach against all validation targets, is the highest among the 16 models tested. The model forms a basis for future development and improvement for fuels containing ammonia blended with hydrocarbons. Sensitivity and reaction path flux analyses were performed to identify the important reactions controlling global reactivity in addition to reactant and product consumption and formation.

At high temperatures ($T > 1500$ K), fuel reactivity for pure NH_3 and NH_3/H_2 mixtures is found to be sensitive to fuel initiation reactions, including $\dot{\text{N}}\text{H}_2$ addition reactions to NO, and the formation and consumption reactions associated with N_2H_2 and HNO. Among these, the reactions involving $\dot{\text{H}}$ atoms are of particular importance, because $\dot{\text{H}}$ atoms are the key to chain branching via $\dot{\text{H}} + \text{O}_2 = \dot{\text{O}} + \dot{\text{O}}\text{H}$ at high temperatures.

At low to intermediate temperatures, the key reactivity-controlling reactions shift to the reactions of $\dot{\text{N}}\text{H}_2$ with HO_2 , NO_2 , NO and O_2 because of the abundance of HO_2 at lower temperatures. Moreover, the important H_2NO radical intermediate species undergoes H-atom abstraction reactions, further dictating the fuel reactivity through the competition between the chain-propagation and chain-termination reaction pathways. The formation and consumption pathways of NO, NO_2 and N_2O , are highly coupled with the important reactions governing fuel reactivity.

In terms of fundamental combustion experiments of NH_3/H_2 , additional RCM IDT data would be beneficial for cross-checking the existing data, which show some deviations at overlapping conditions, and would further improve the predictions of the kinetic model at low temperatures. Moreover, more laminar flame speed and speciation data measurements at pressures higher than 1 atm would also be beneficial for model improvement. High-level ab initio calculations and direct measurements of the rate constants of $\dot{\text{N}}\text{H}_2$ reactions with $\dot{\text{O}}$, and N_2H_2 unimolecular decomposition reactions is desirable to further reduce the uncertainty in the model, particularly in predicting flame conditions.

Novelty and significance statement

In this study, we elucidate the combustion chemistry of NH_3 and its blends with H_2 in detail and propose a new chemical kinetic model which can accurately capture most fundamental combustion experimental data of NH_3/H_2 available in the literature, including ignition delay times, laminar flame speeds and species concentration profiles. The high-fidelity chemical kinetic model developed herein forms a basis for future development and improvement for fuels containing ammonia blended with hydrogen and hydrocarbons.

CRedit authorship contribution statement

Yuxiang Zhu: Investigation, Writing – original draft. **Henry J. Curran:** Resources, Writing – review & editing, Supervision, Project administration. **Sanket Girhe:** Resources. **Yuki Murakami:** Resources. **Heinz Pitsch:** Supervision, Project administration. **Kelly Senecal:** Project administration. **Lijun Yang:** Project administration. **Chong-Wen Zhou:** Resources, Writing – review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Acknowledgments

We would like to thank Prof. William J. Pitz for helpful discussions. The computational resources for this work were provided by the High-performance Computing (HPC) center at Beihang University. Yuxiang Zhu acknowledges support from the International Joint Doctoral Education Fund of Beihang University. The work of Sanket Girhe is supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy - Cluster of Excellence 2186 "The Fuel Science Center" - ID: 390919832.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2023.113239.

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